User Experience

FINTECH 512
Software Engineering

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Outline

- What is 'User Experience?'
- Elements of user experience
- Human characteristics
  - Memory
  - Attention
  - Perception
    - Gestalt principles
  - Pointing!
- Accommodating human characteristics
  - Mapping
  - Feedback
  - Constraints
- Usability Heuristics
- Measuring Usability

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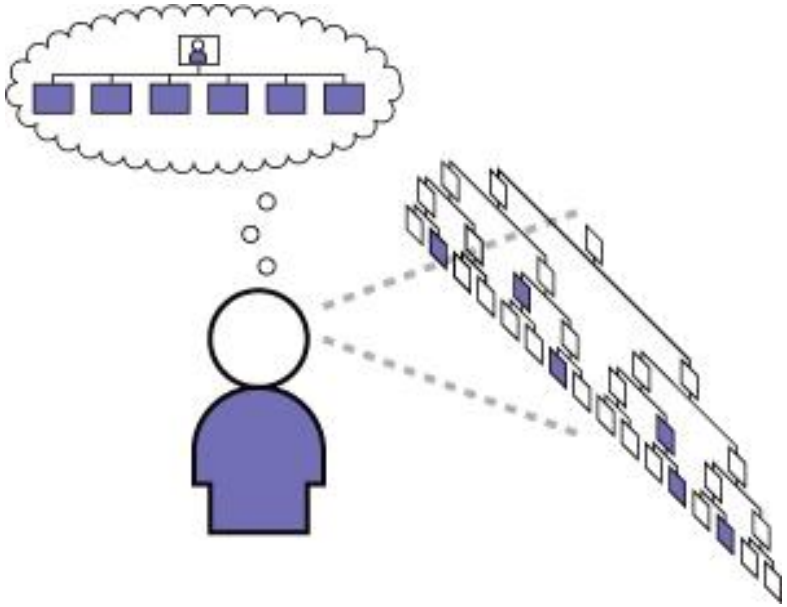
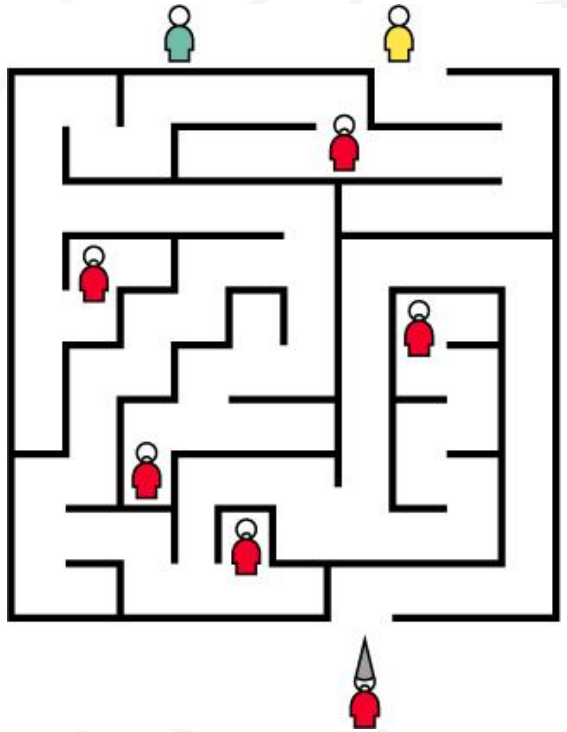
User Experience

- “the experience [a] product creates for the people who use it in the real world”





'The Elements of User Experience: User-Centered Design for the Web and Beyond’ – Jesse James Garrett

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User Experience (UX)

- An aspect of the 'usability' quality attribute
- ”User interface design should ensure that interaction between the human and the machine provides for effective operation” [0]
- “Usability principles can provide guidelines for discovering problems in the design of the user interface” [0]
- “If a software system is to achieve its full potential, it is essential that its user interface should be designed to match the skills, experience and expectations of its anticipated users.“ – Ian Sommerville [1]

[0] SWEBOK
[1] https://ifs.host.cs.st-andrews.ac.uk/Books/SE9/WebChapters/PDF/Ch_29%20Interaction_design.pdf

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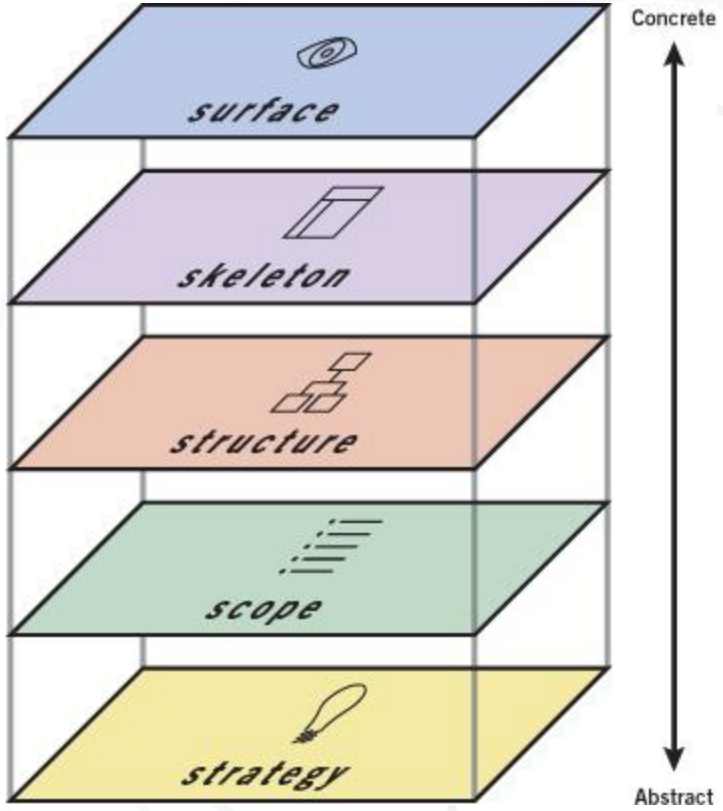
UX Elements: Five Layers(‘Planes’)

- Surface – images, text, input controls
- Skeleton – placement of images, text, controls (e.g. on a page)
- Structure – navigation (e.g. between pages)
- Scope – features and functions of the product
- Strategy – what the users and operators of the product hope to accomplish



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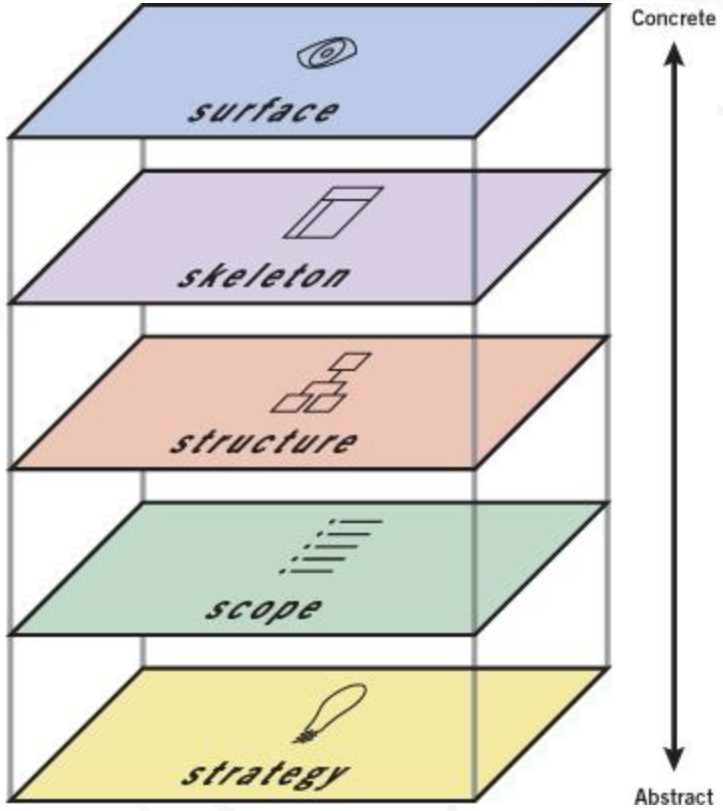
UX Elements: Five Layers(‘Planes’)

- Surface – images, text, input controls (UX Design)
- Skeleton – placement of images, text, controls (Use Cases, UX Design)
- Structure – organization, navigation (Use Cases)
- Scope – features and functions of the product (Requirements)
- Strategy – what the users and operators of the product hope to accomplish



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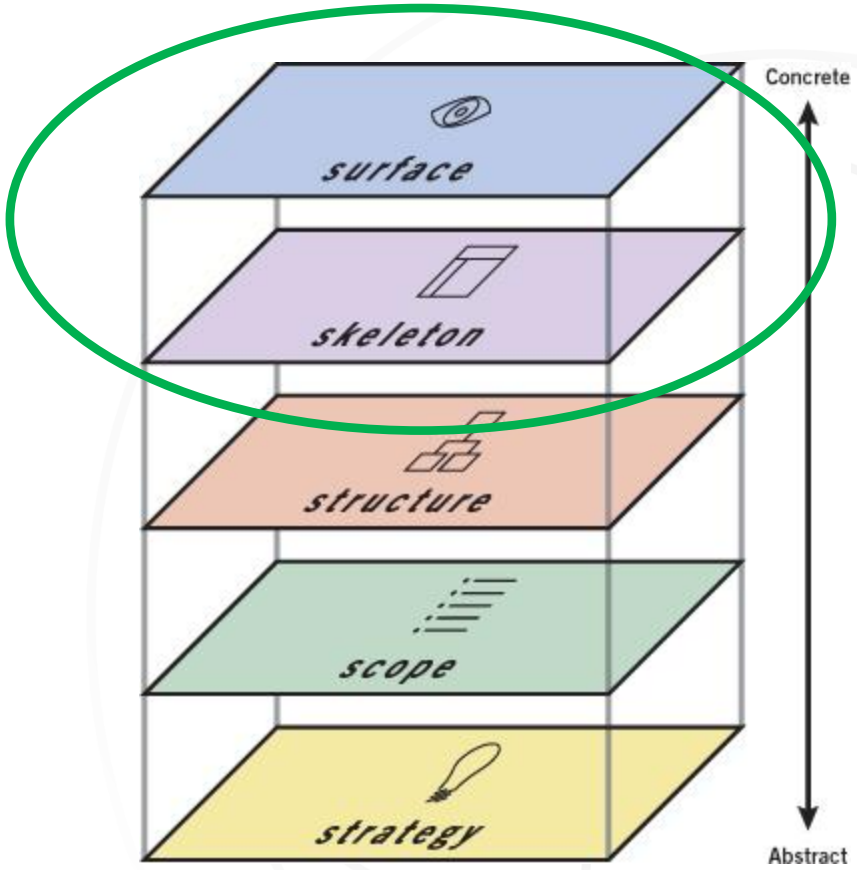
Today: Some principles underlying the top layers

- Surface – images, text, input controls (UX Design)
- Skeleton – placement of images, text, controls (UX Design)
- Structure – organization, navigation (Use Cases)
- Scope – features and functions of the product (Requirements)
- Strategy – what the users and operators of the product hope to accomplish



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Human Characteristics

- Memory
- Attention
- Perception
  - Gestalt principles
- Pointing

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Short/Long-term Memory & Attention

Memory

Short-term

- Limited capacity (~7 items)
- Instant recall
- Fragile

Long-term

- Unlimited capacity
- Slower recall
- Needs organization

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Memory Implications

Minimize	need for mighty memory
Present	information structured, familiar, meaningful ways
Place	all required information for task performance in close physical proximity
Give	the user control over the pace of information presentation

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Attention

Concentration of the mind on a single object or thought.

Attention Implications

ATTENTIONS IS PRECIOUS AND LIMITED

DON’T LITTER YOUR INTERFACE WITH DISTRACTING MATERIAL

FOCUS USERS’ ATTENTION ON IMPORTANT ELEMENTS

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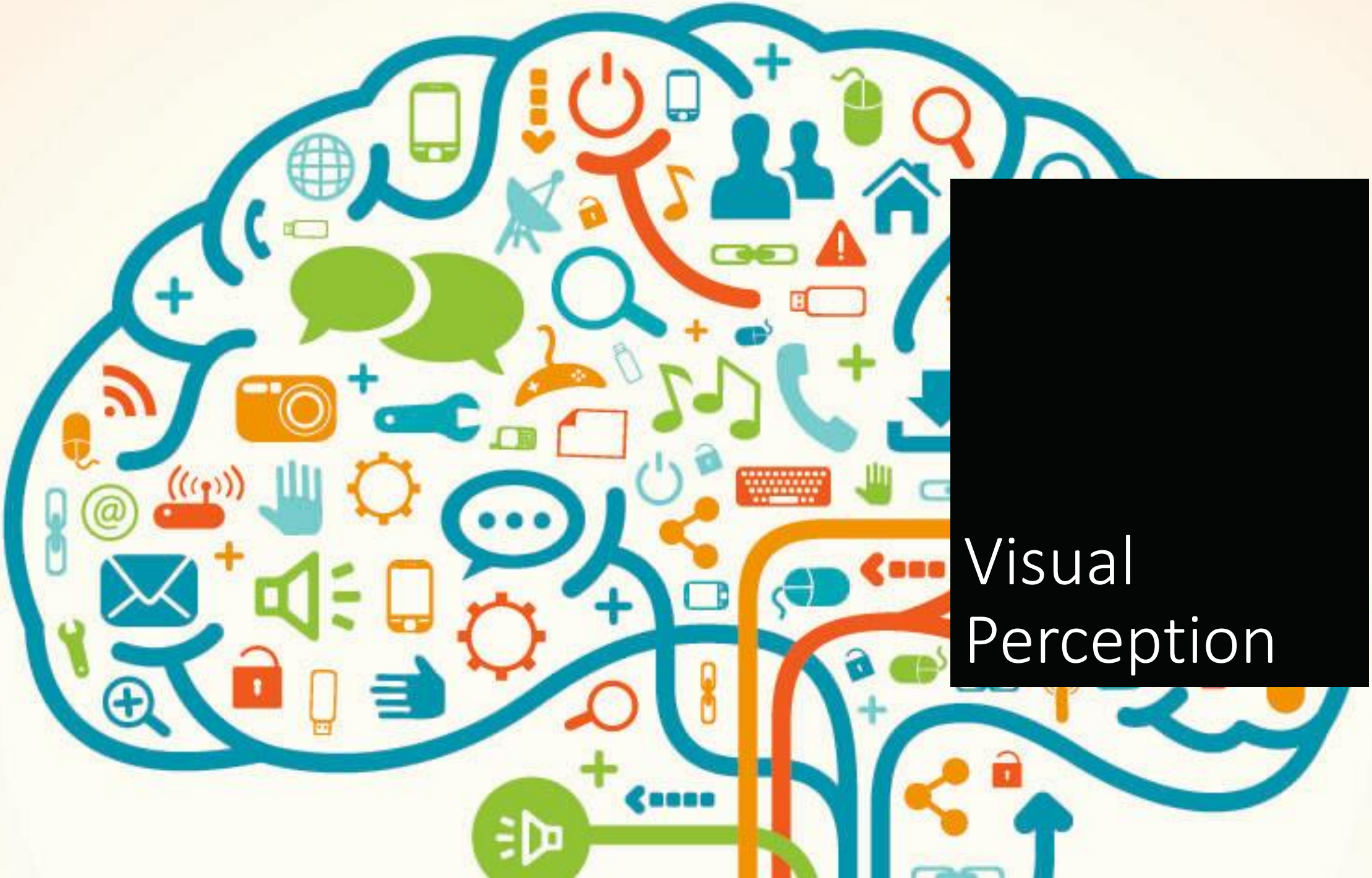

Visual Perception

Gestalt Principles

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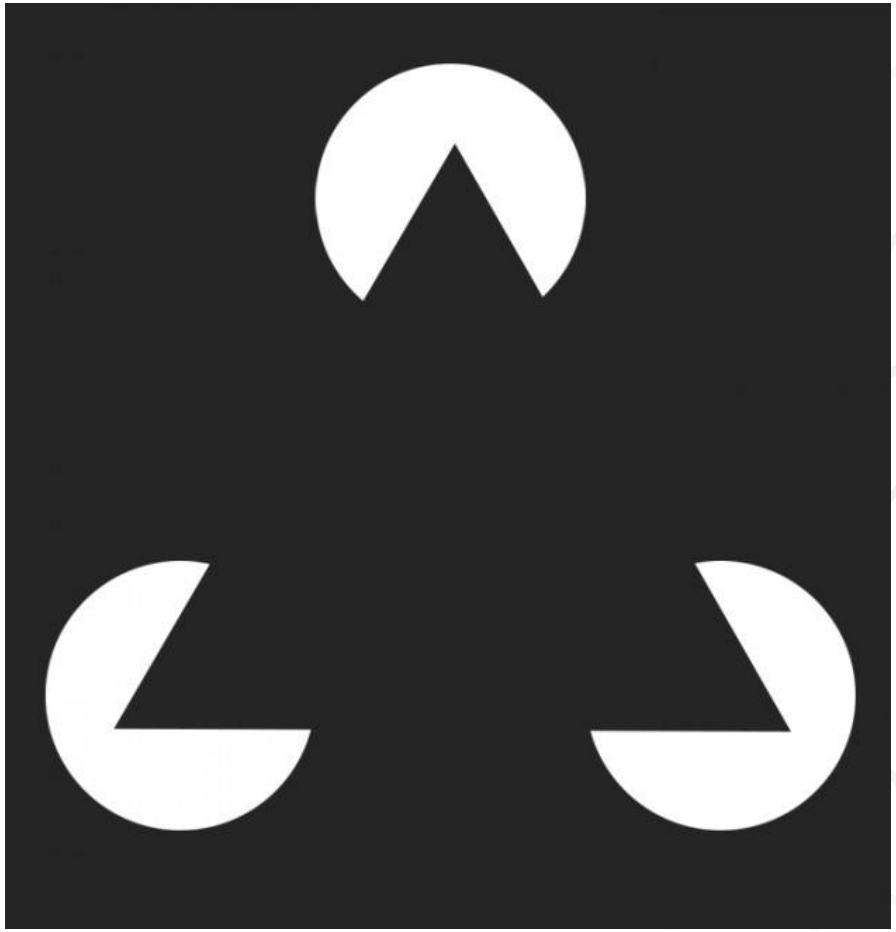
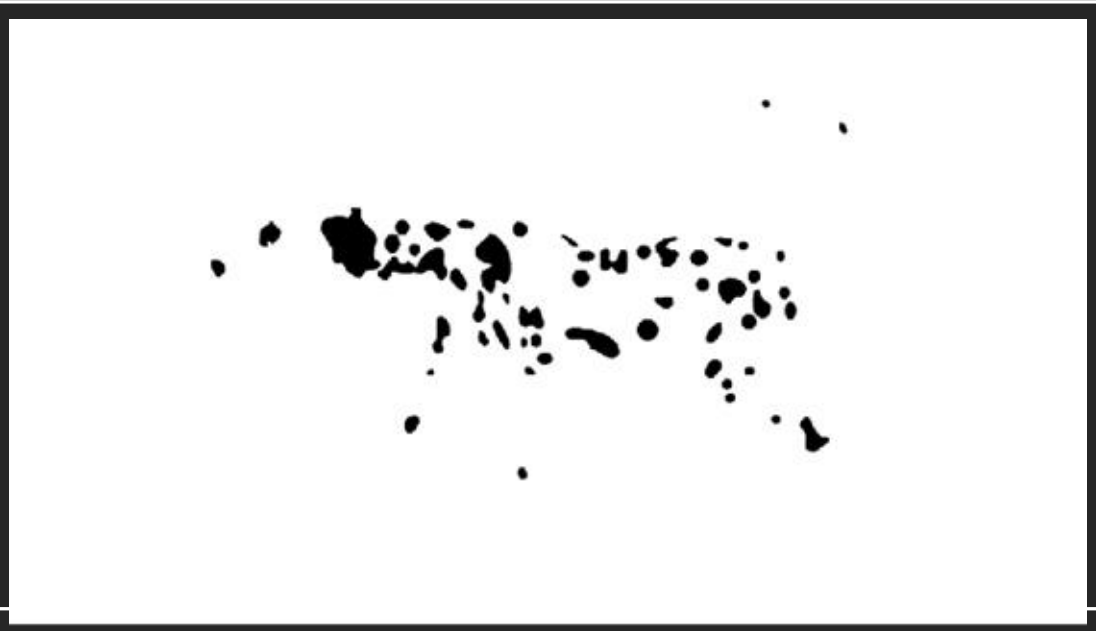
What do you see?



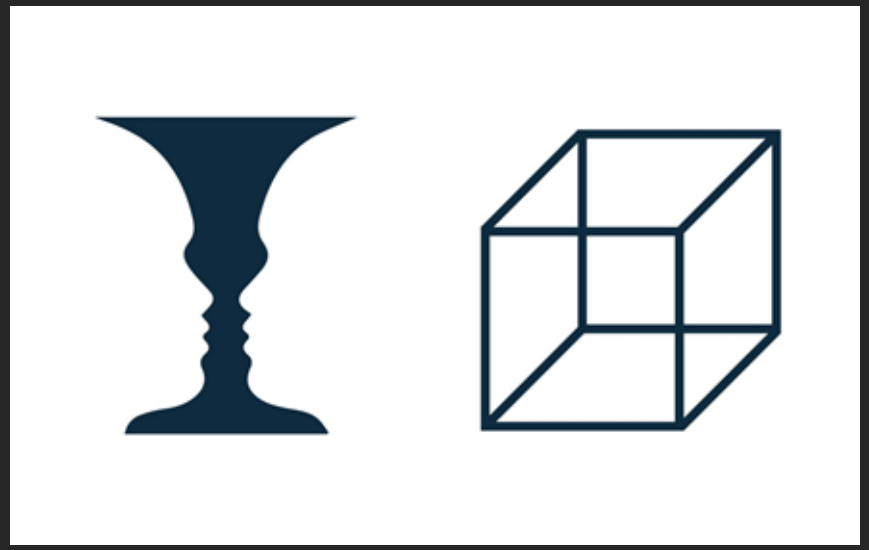
An organized whole, is perceived as greater than the sum of its parts

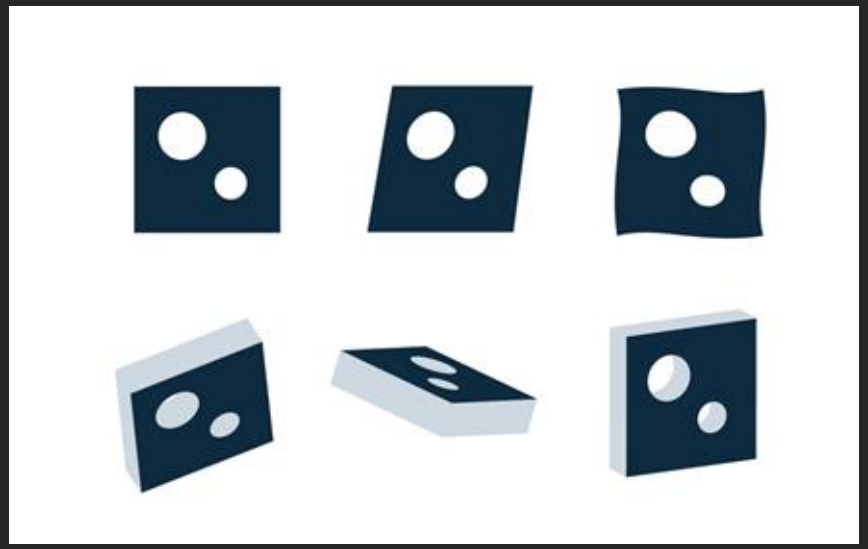
Gestalt Principles of Visual Perception

Emergence



Multi-stability





Invariance

Reification

#1 Figure-Ground

The figure-ground principle states that people instinctively **perceive objects as either being in the foreground or the background**. They either stand out prominently in the front (the figure) or recede into the back (the ground).

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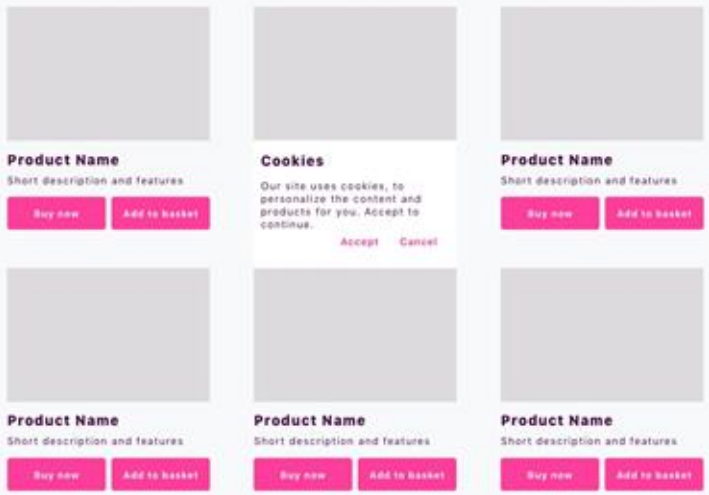
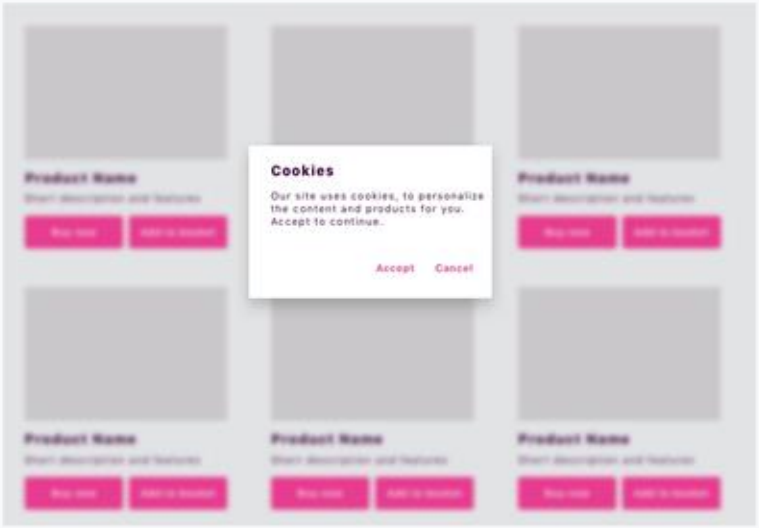
Figure-Ground

Figure - Ground

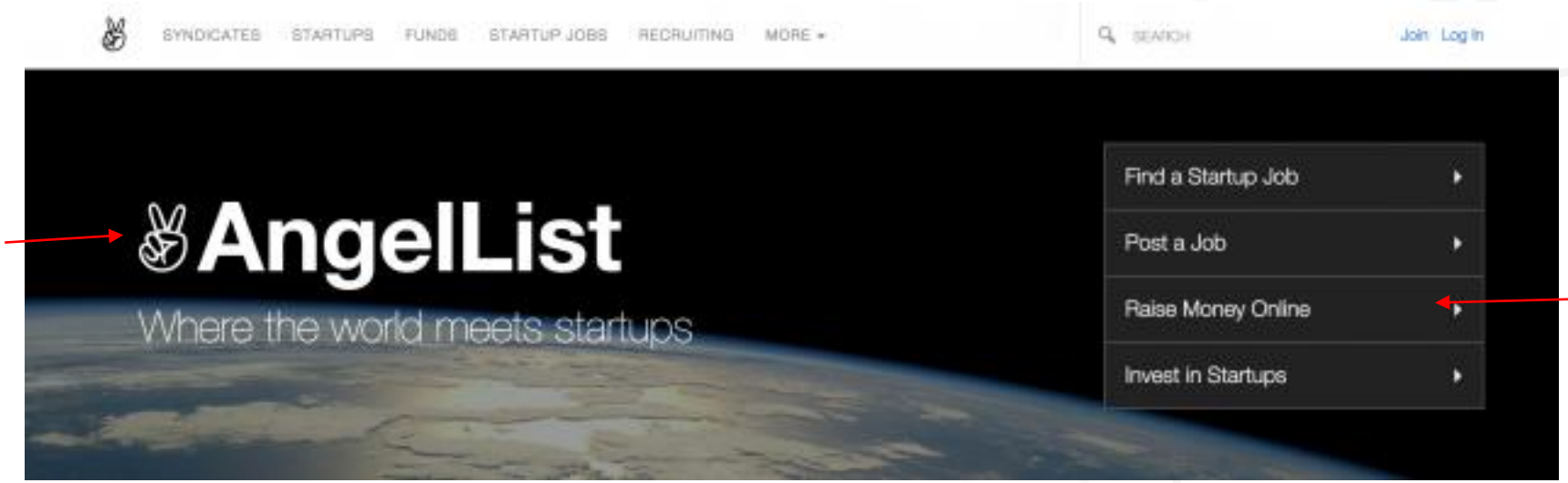




Gestalt Principles

https://uxmisfit.com

Logo



White text on black background

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#2 Similarity

The principle of similarity states that **when things appear to be similar** to each other, **we group them** together. And we also tend to think they have the same function.

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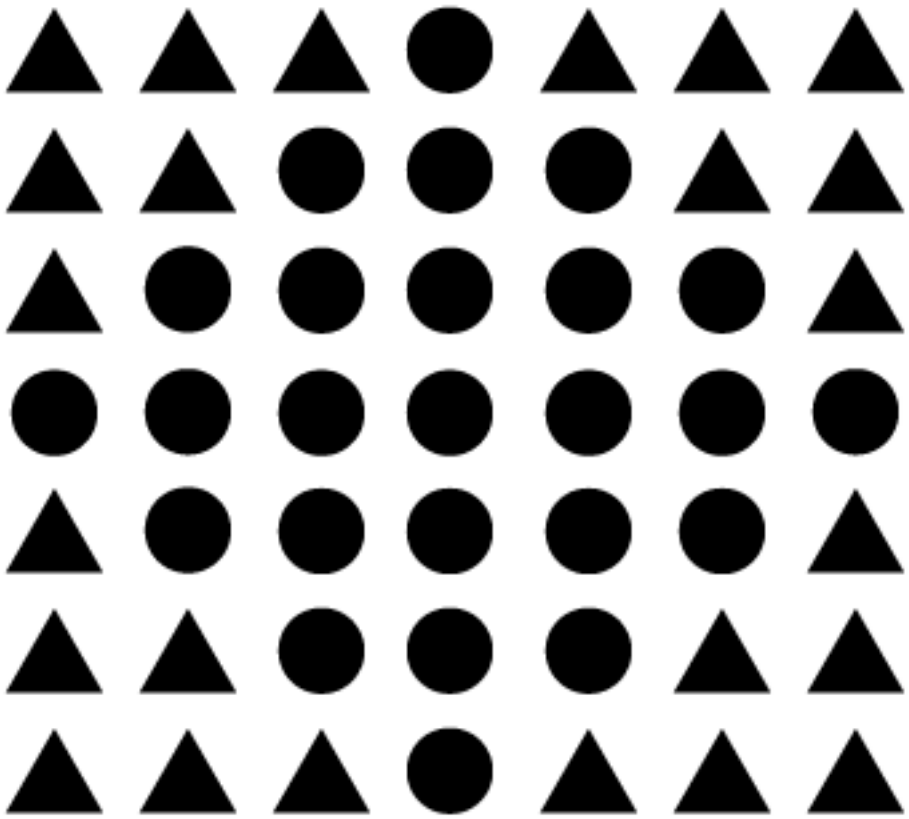
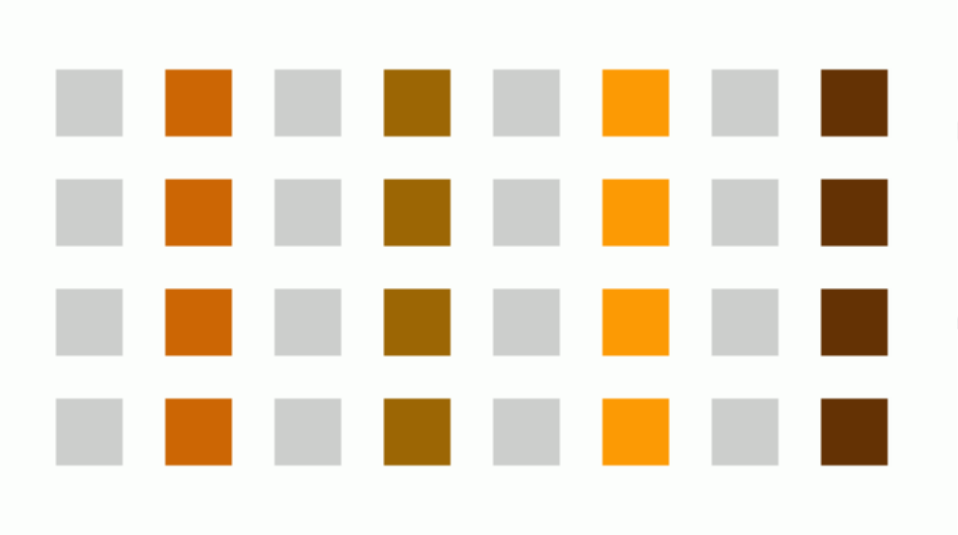
Similarity





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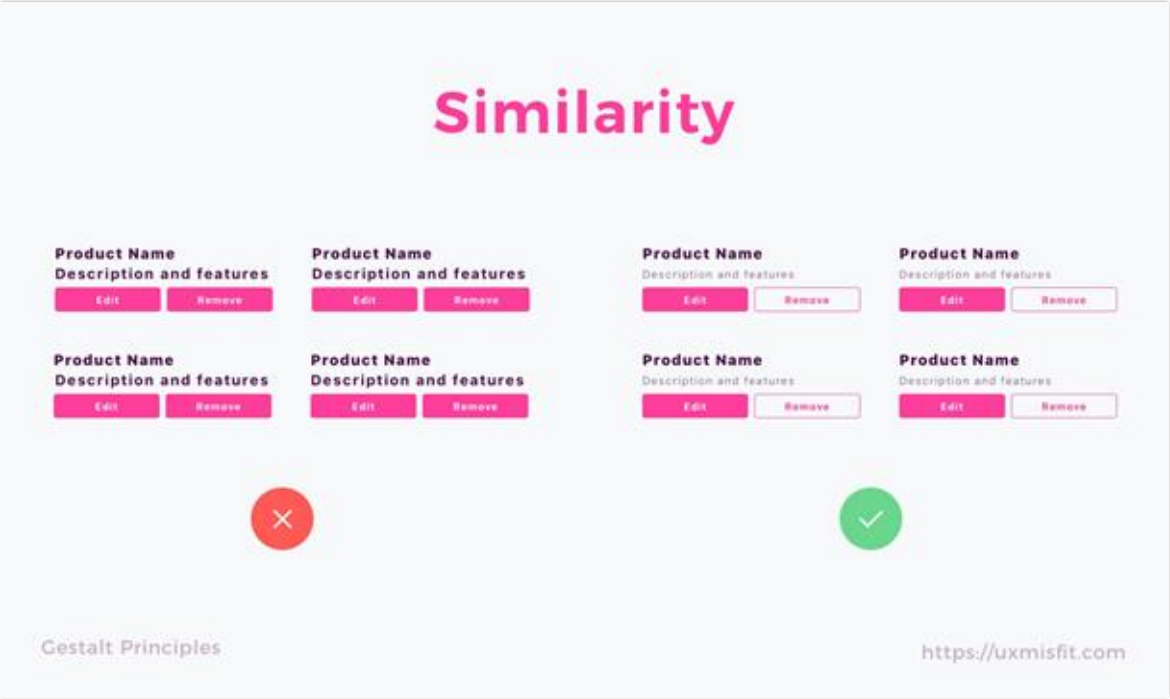
Similarity
Product Name
Description and features
Edit
Remove
Product Name
Description and features
Edit
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Product Name
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Gestalt Principles
https://uxmisfit.com

Similarity

Gray

Black

Blue links

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#3 Proximity

The principle of proximity states that things that are **close together** appear to be **more related** than things that are spaced farther apart.

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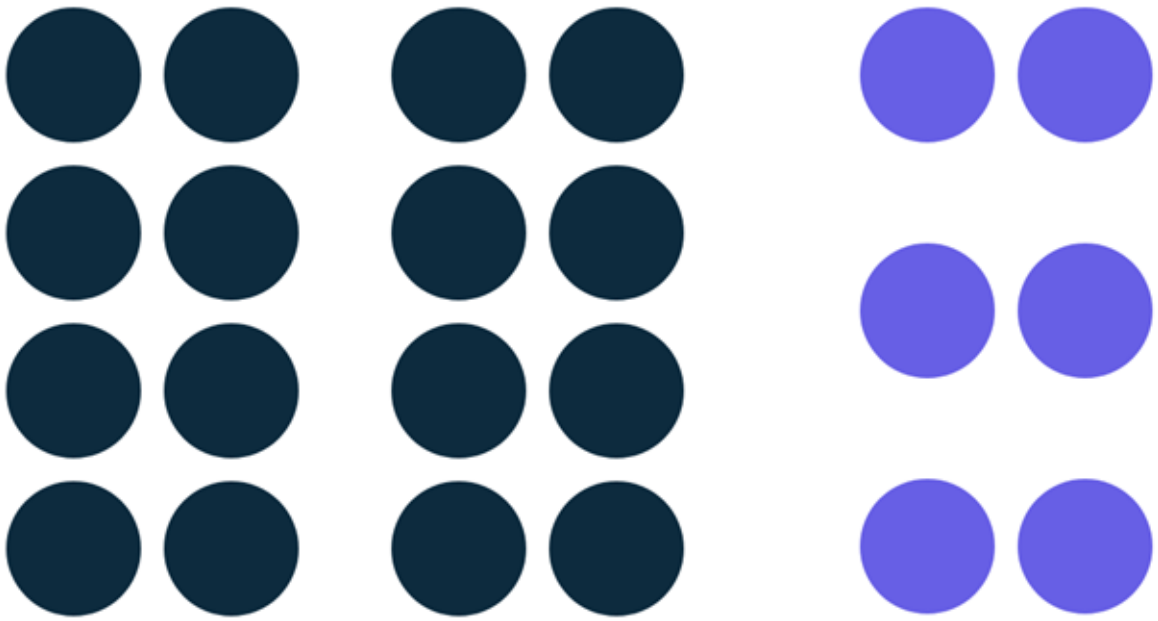
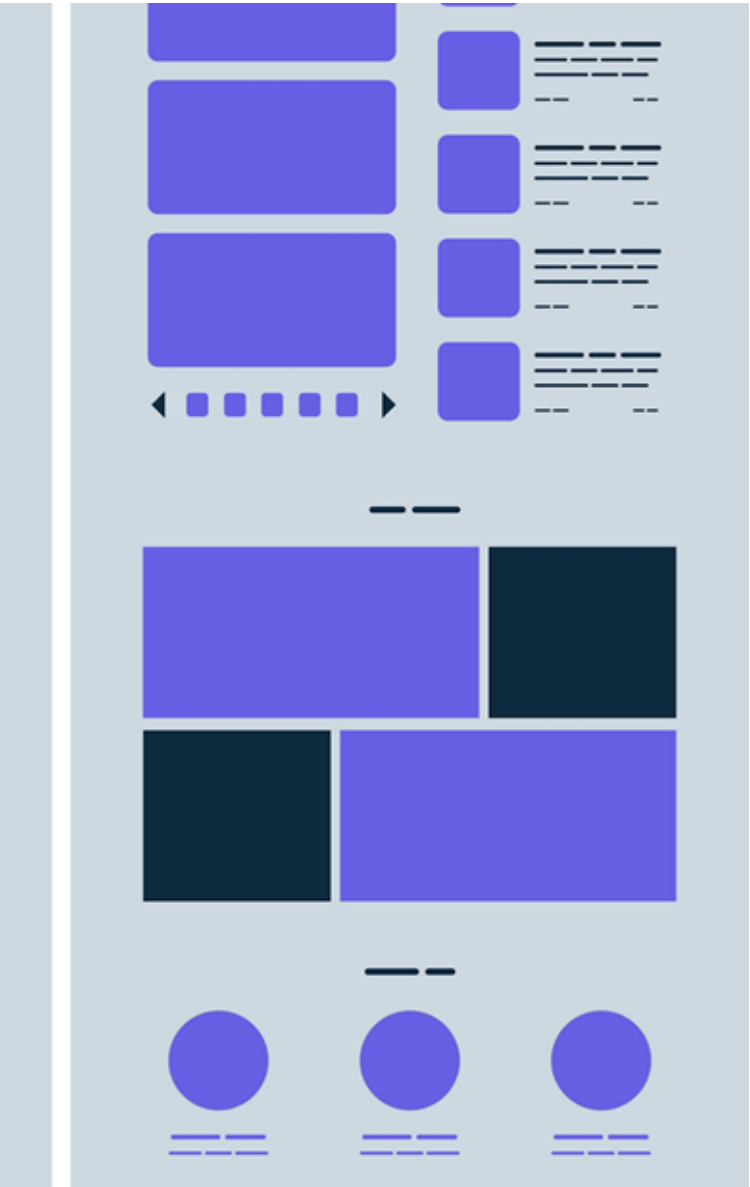
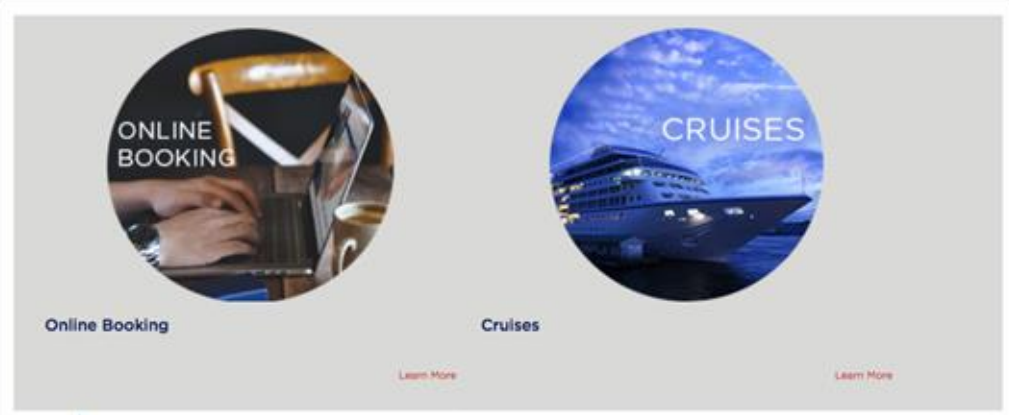
Proximity



Proximity



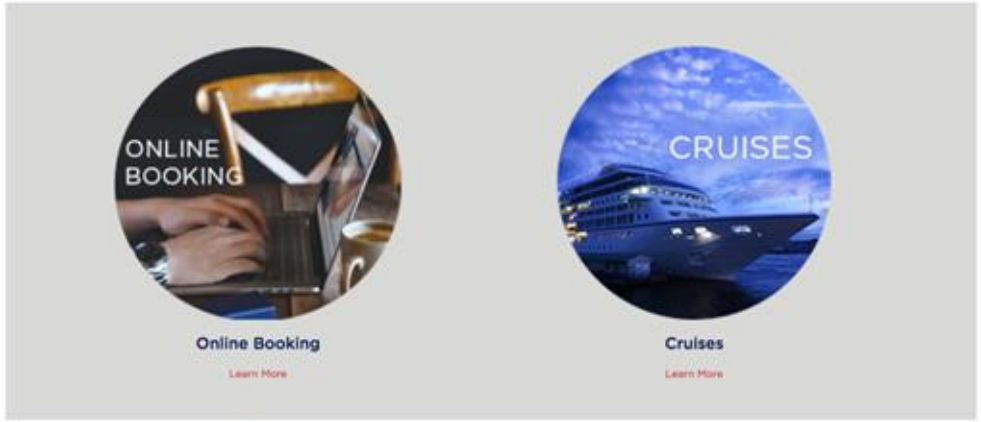
Design Problem



Category Title (left aligned)

Category link (right aligned)

Design solution using Proximity



Category Title & Link (center aligned)

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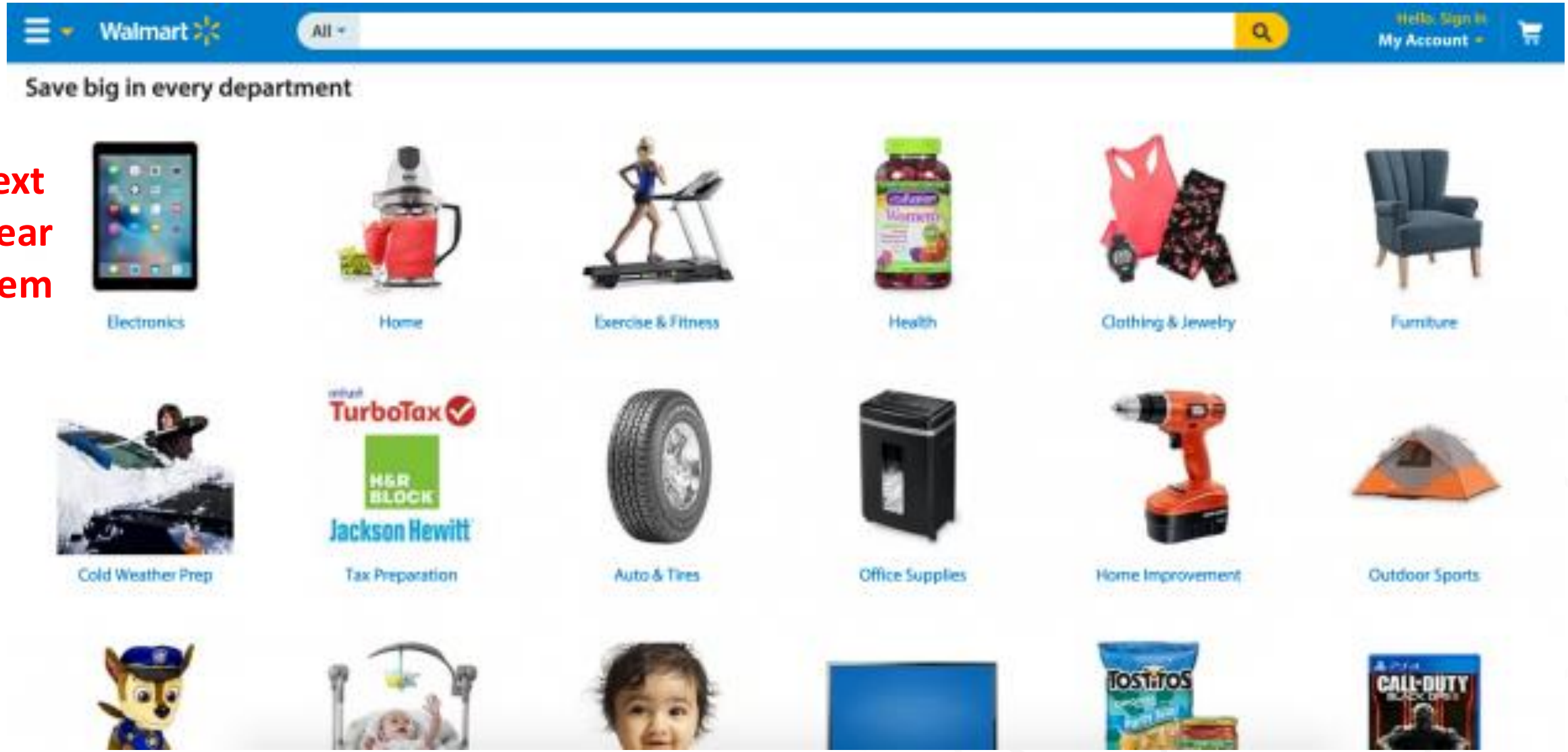
Text near item



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#4 Common Region

The principle of common region is highly related to proximity. It states that when objects are located within the **same closed region**, we perceive them as being **grouped together**.

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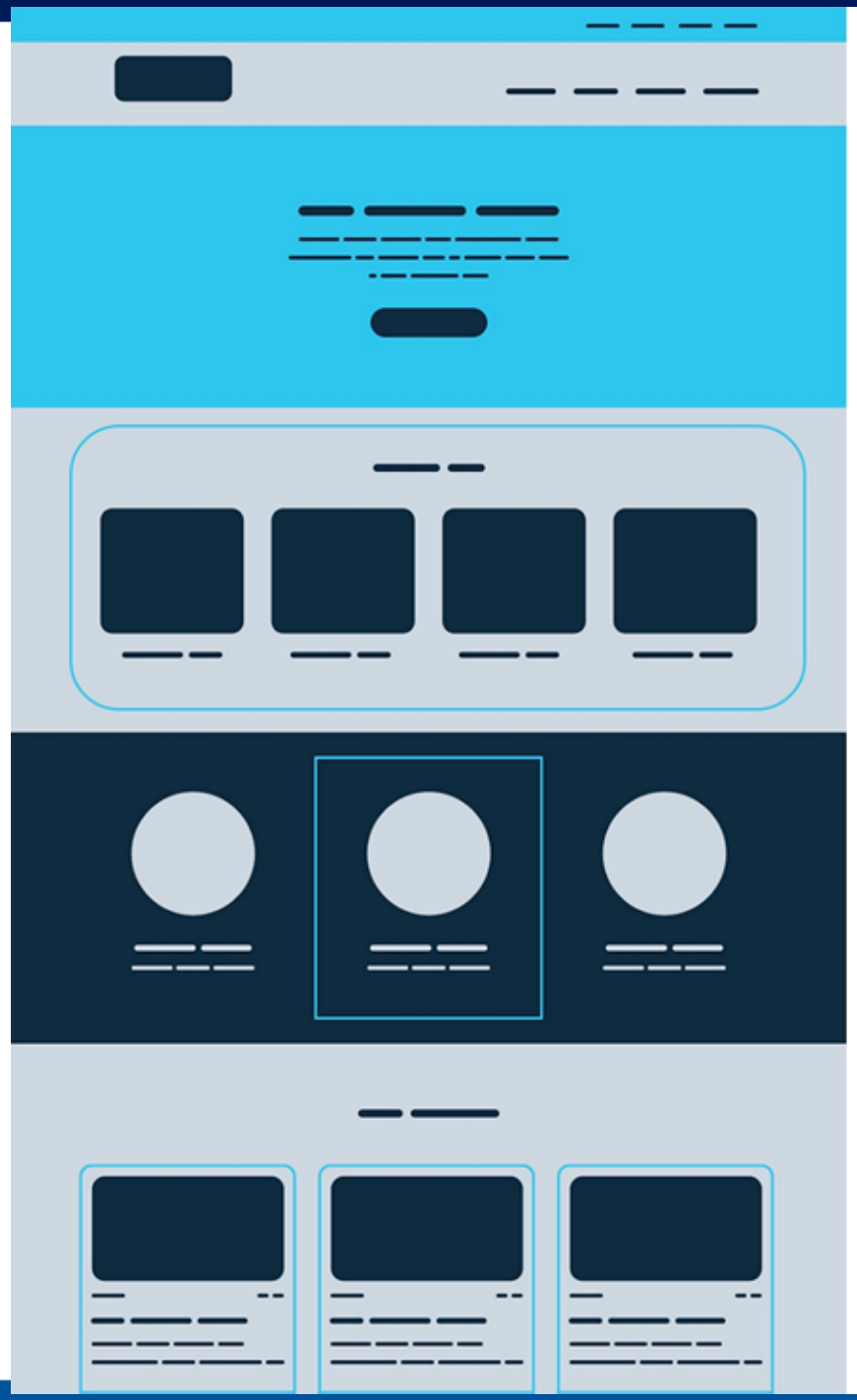

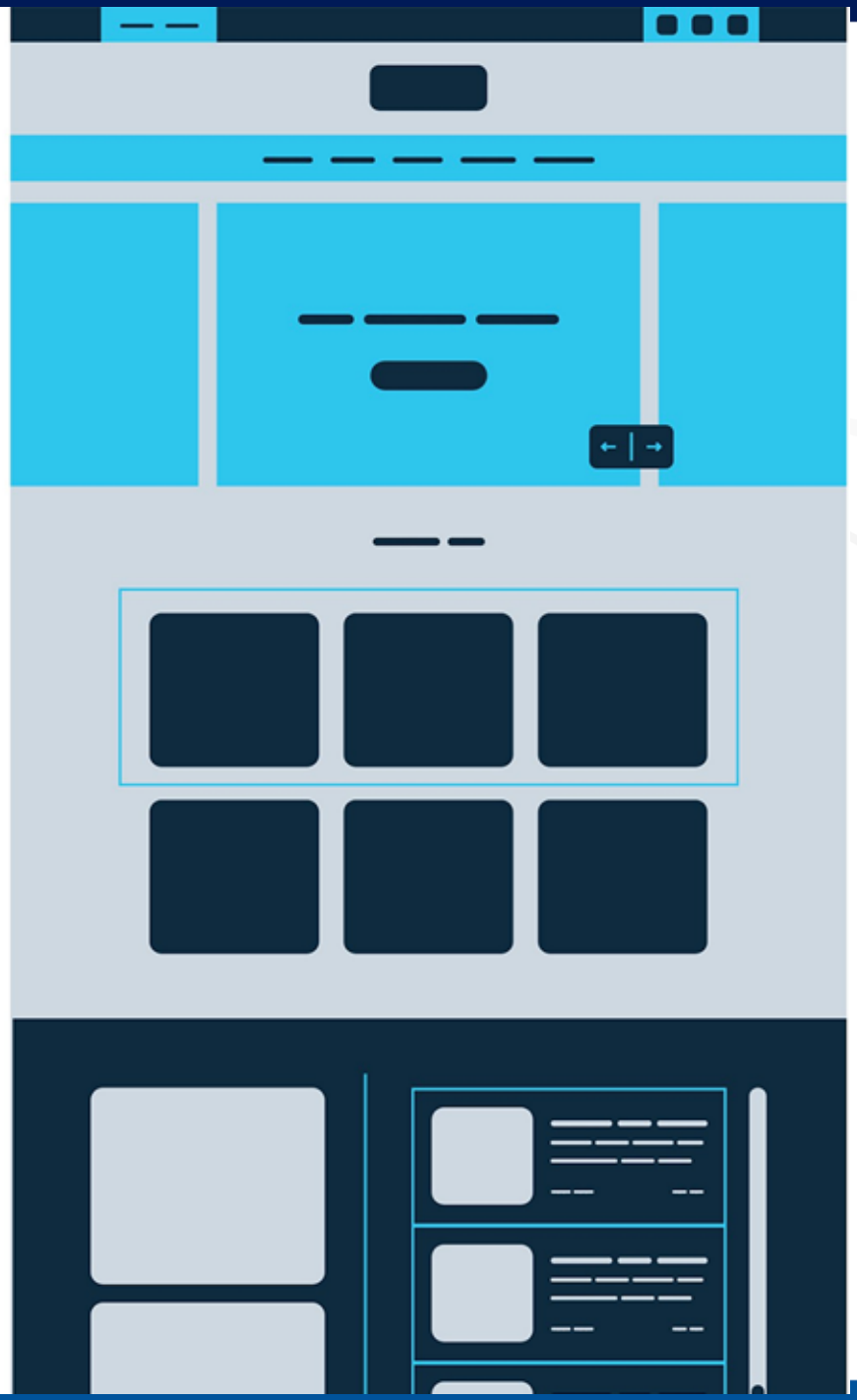

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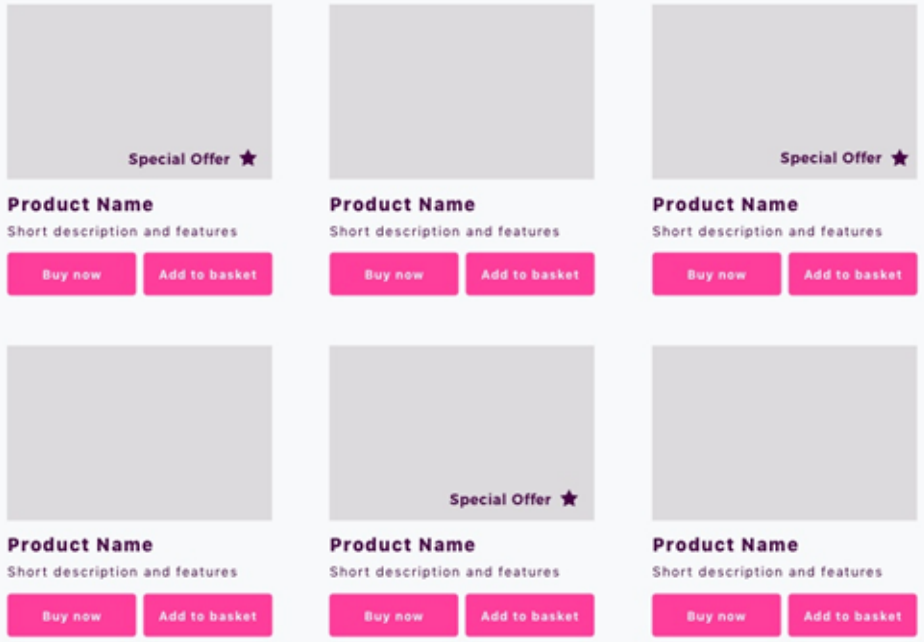
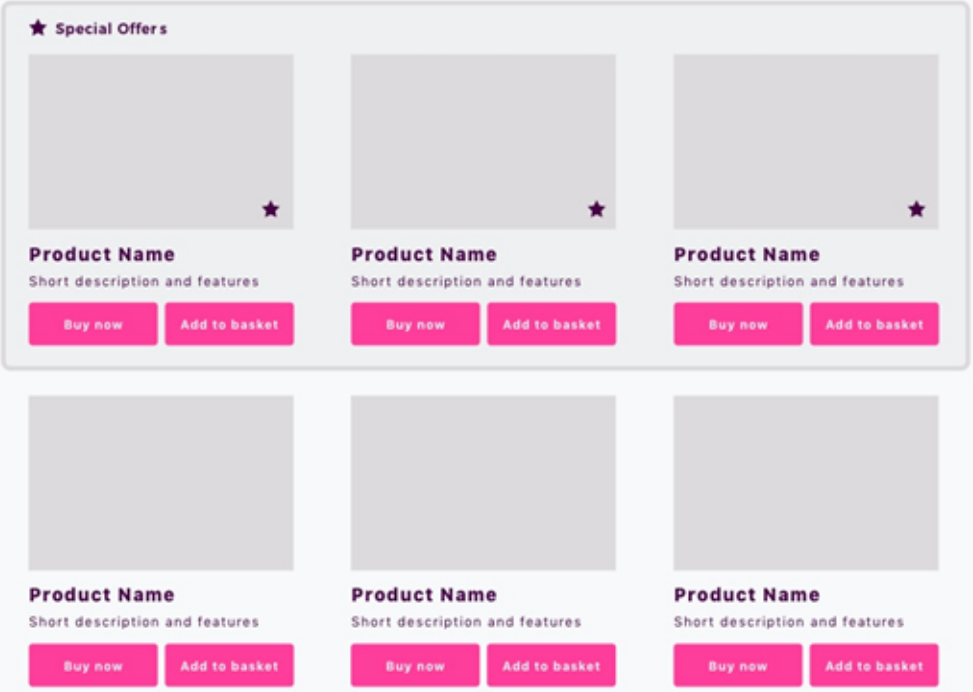
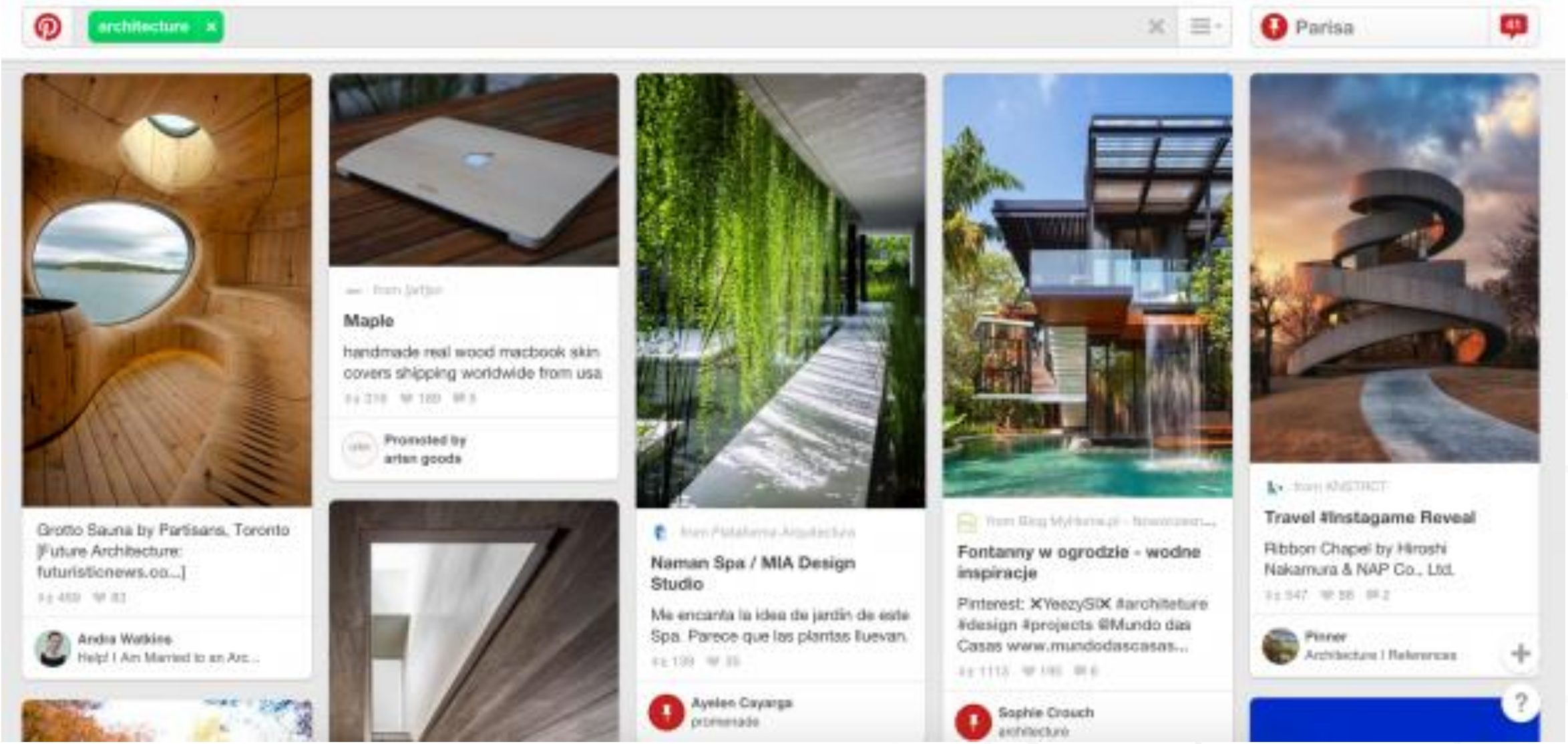
Common Region






architecture
Parisa
Grotto Sauna by Partisans, Toronto [Future Architecture: futuristicnews.co...]
Andra Watkins
Help! I Am Married to an Arc...
Maple
handmade real wood macbook skin covers shipping worldwide from usa
Promoted by
artsn goods
Naman Spa / MIA Design Studio
Me encanta la idea de jardín de este Spa. Parece que las plantas llevan.
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Saphie Crouch
Travel #Instagame Reveal
Ribbon Chapel by Hiroshi Nakamura & NAP Co., Ltd.
Pinner
Architecture | References

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#5 Continuity

The principle of continuity states that elements that are **arranged on a line or curve** are perceived to be **more related** than elements not on the line or curve.

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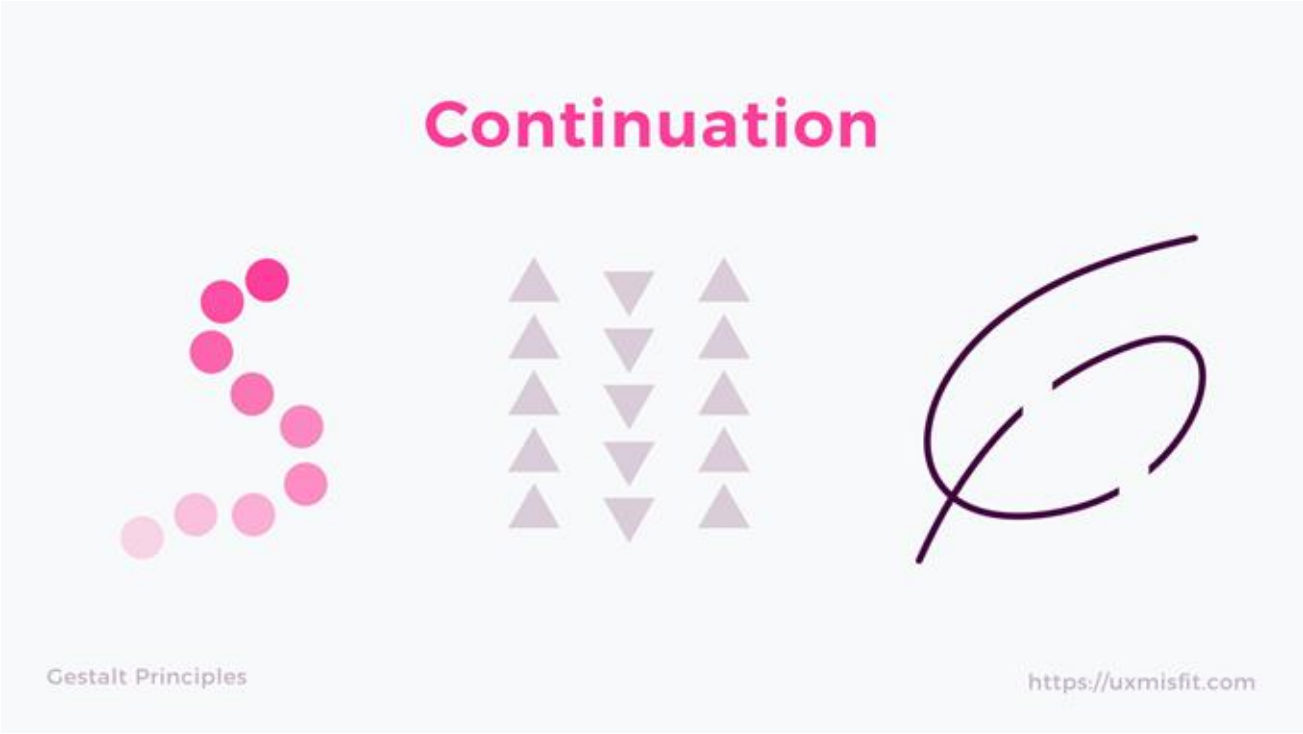
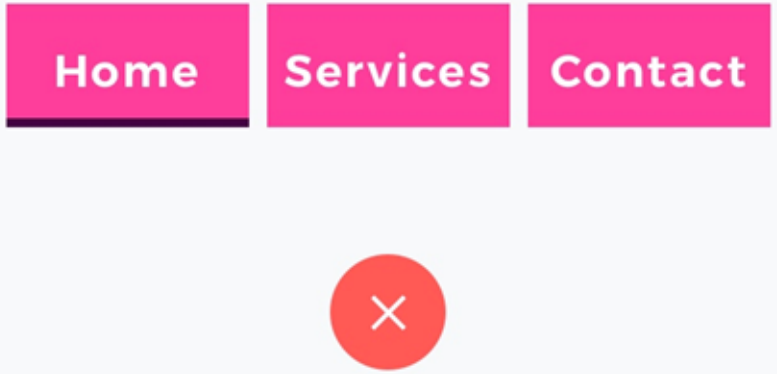
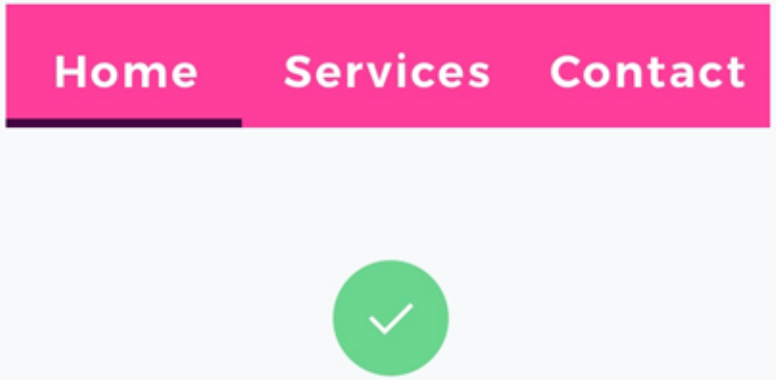
Continuity/ Continuation



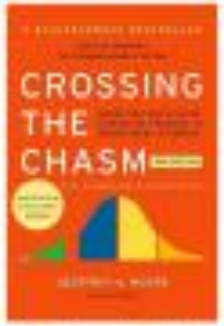
Continuation





Customers Who Bought This Item Also Bought

Page 1 of 20



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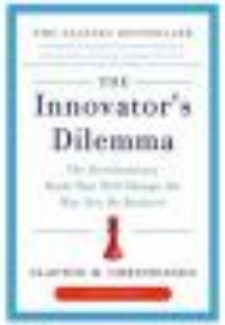
The Lean Startup: How Today's Entrepreneurs Use Continuous Innovation to
› Eric Ries
1,062
Hardcover
$16.66 Prime



The Innovator's Dilemma: The Revolutionary Book That Will Change the Way
› Clayton M. Christensen
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#1 Best Seller in Industrial Management...
Paperback
$10.06 Prime

The Innovator's Solution: Creating and Sustaining Successful Growth
› Clayton M. Christensen
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Hardcover
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› Clayton M. Christensen
583
Hardcover
$15.86 Prime

Products in the line are similar or related

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#6 Closure

The principle of closure states that when we look at a complex arrangement of visual elements, **we tend to look** for a single, **recognizable pattern**.

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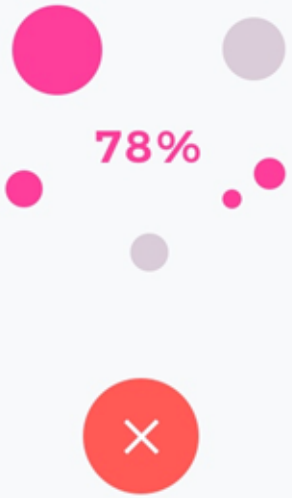
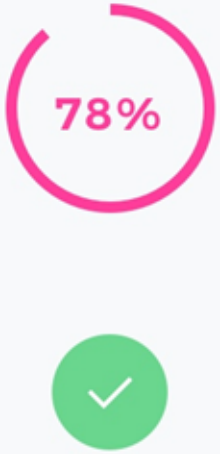
Closure

Closure





#7 Focal Point

The focal point principle states that whatever stands out visually will capture and hold the viewer's attention first.

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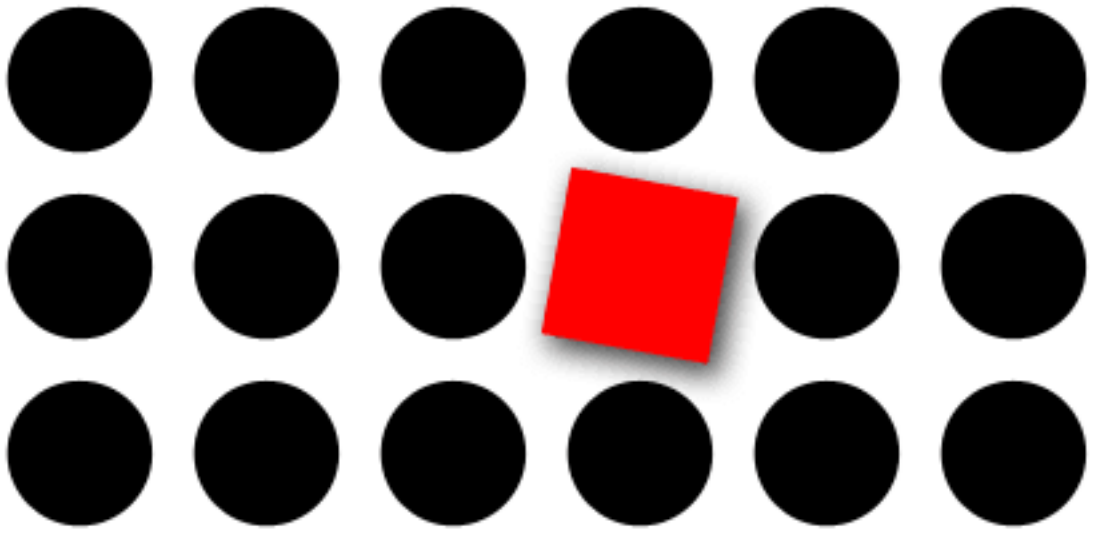

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Call to action

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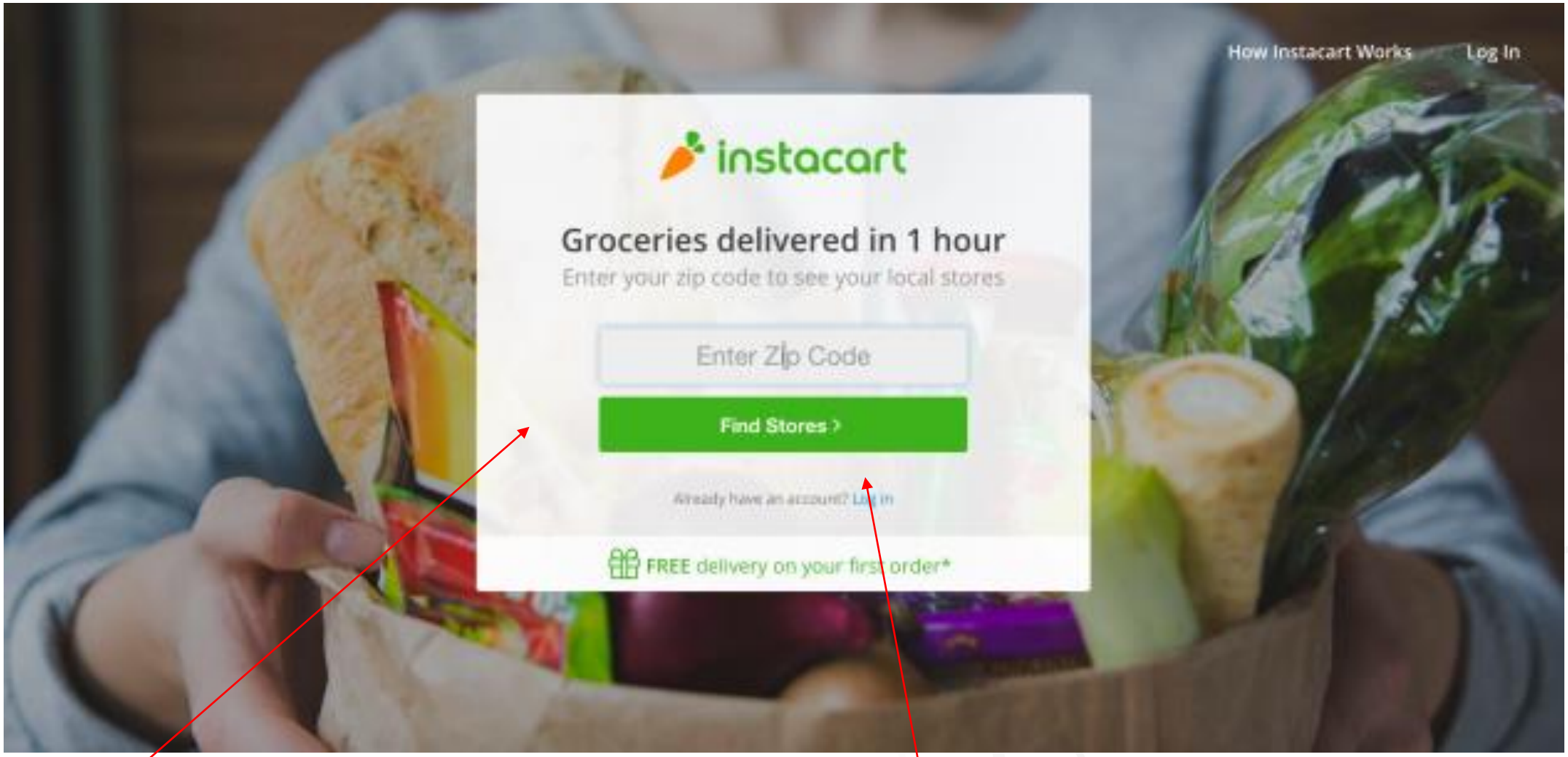

Foreground

Focal: Call to action

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JSON **JSONLint** - The JSON Validator

Try the New Pro

More Developer Tools

1

Validate JSON

Clear

Support JSONLint for $2/Month

https://jsonlint.com/

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Takeaway

Designers should use perceptual capabilities so an interface can be structured in a meaningful and obvious way

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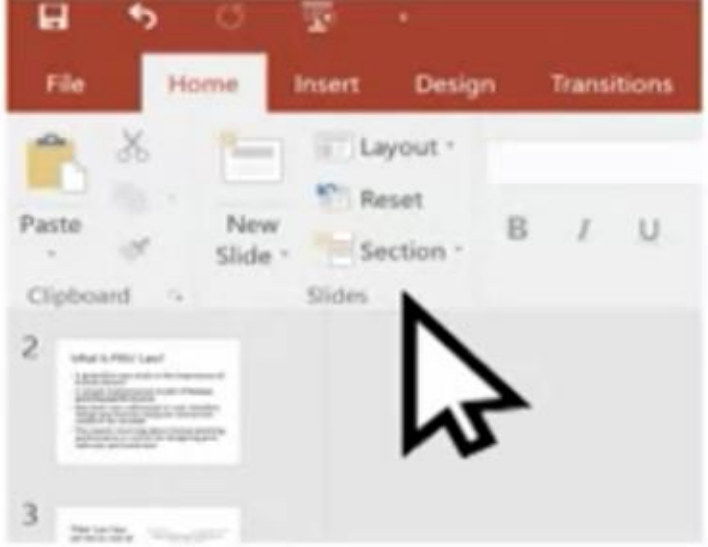
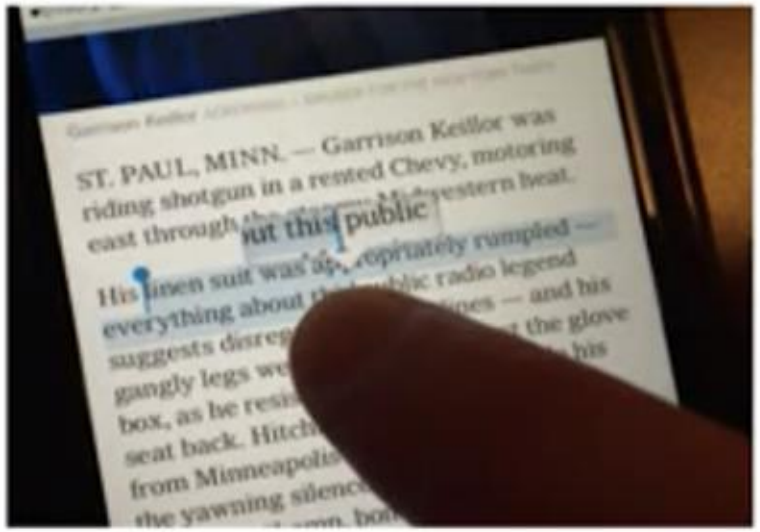


Pointing use the mouse

Pointing use the trackpad



Pointing using your finger!

Pointing is Central

Fitts' Law

$$T = a + b * log_2\left(\frac{A}{W} + 1\right)$$

The amount of time required for a person to move a pointer to a target area is a function of the distance to the target, the size of the target, and the pointing device.

Characteristics of Fitts' Law

Target size Increases, Time Decreases

Target distance Increases, Time Increases

https://uxplanet.org/breaking-down-fitts-law-for-ux-designers-542cabb48f9

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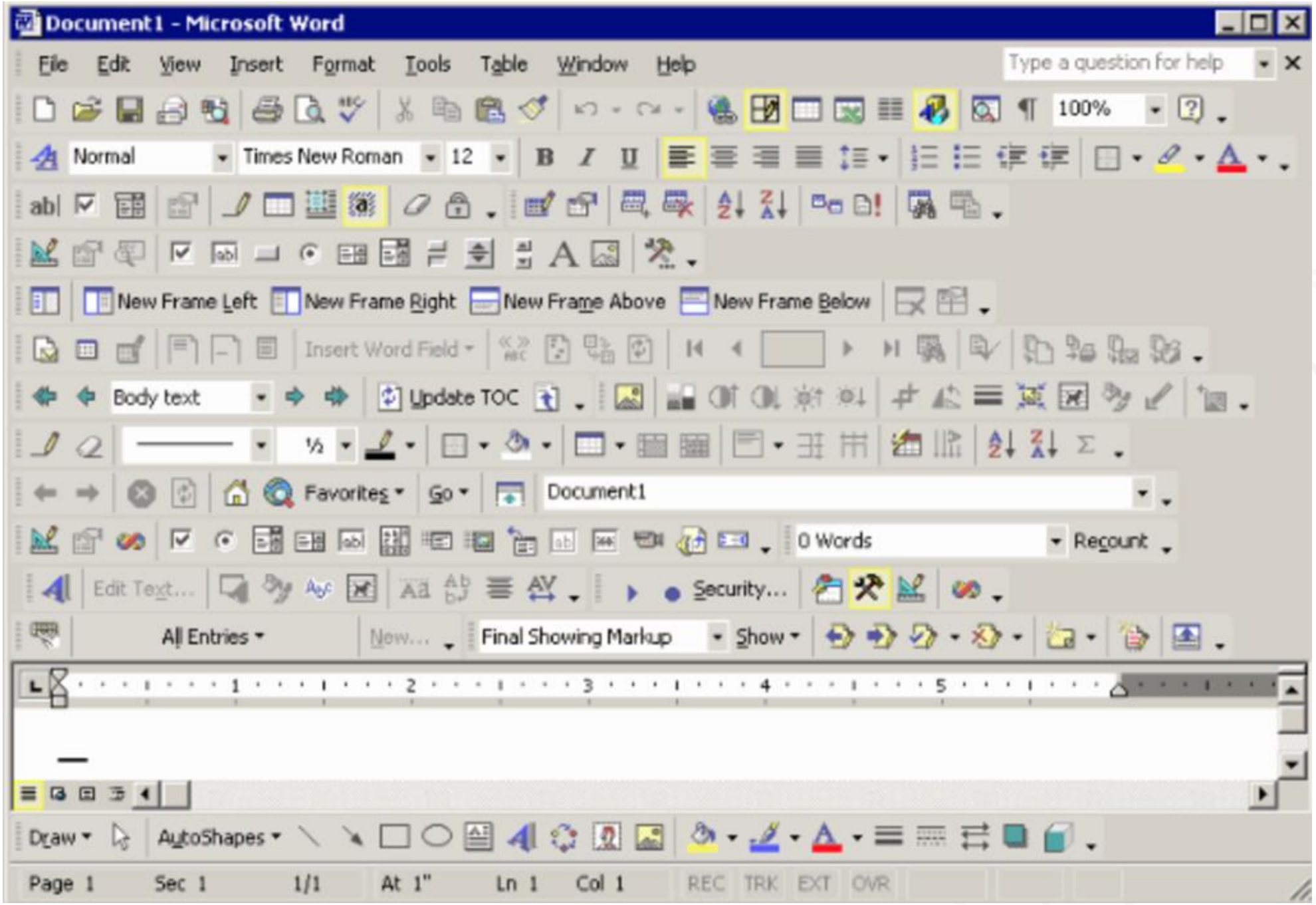
Document1 - Microsoft Word
File Edit View Insert Format Tools Table Window Help
Type a question for help
100%
Normal
Times New Roman
12
New Frame Left
New Frame Right
New Frame Above
New Frame Below
Insert Word Field
Body text
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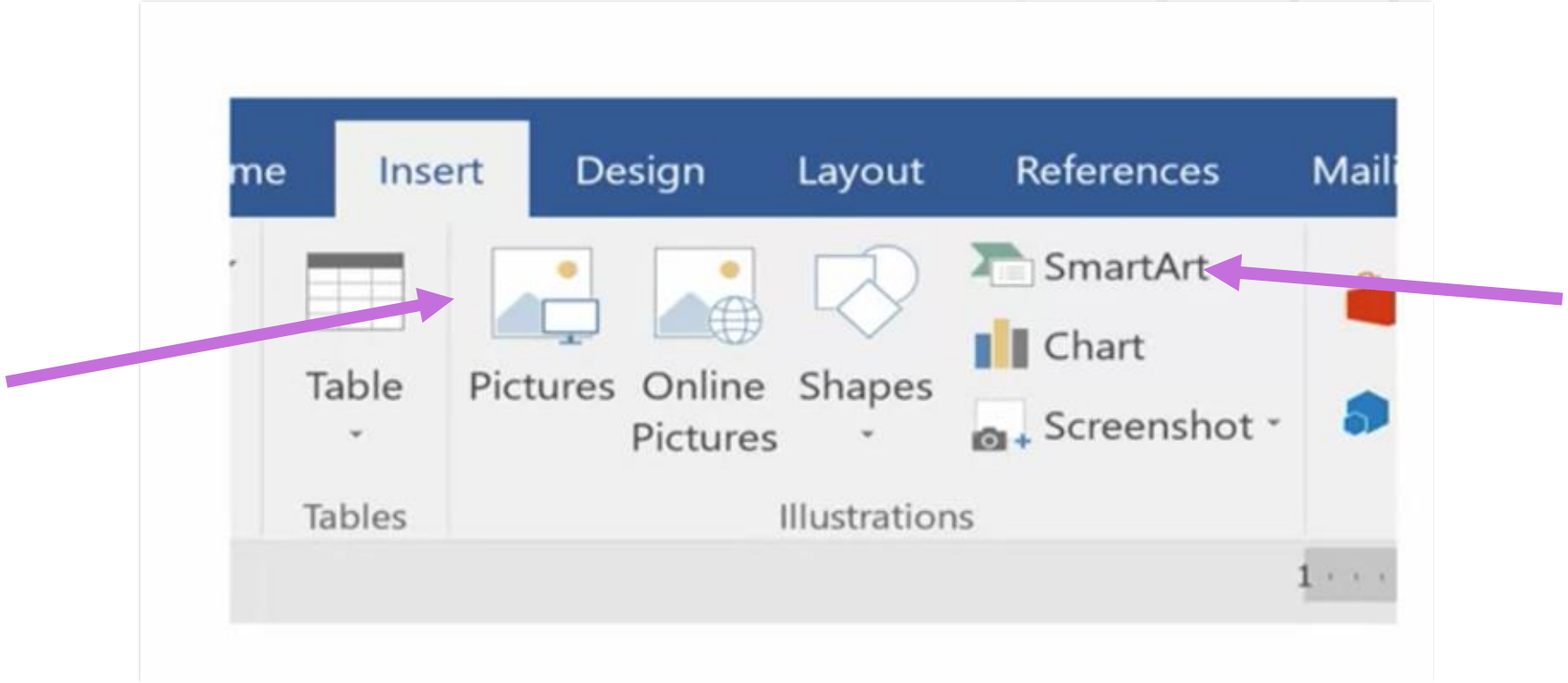
Microsoft Ribbon



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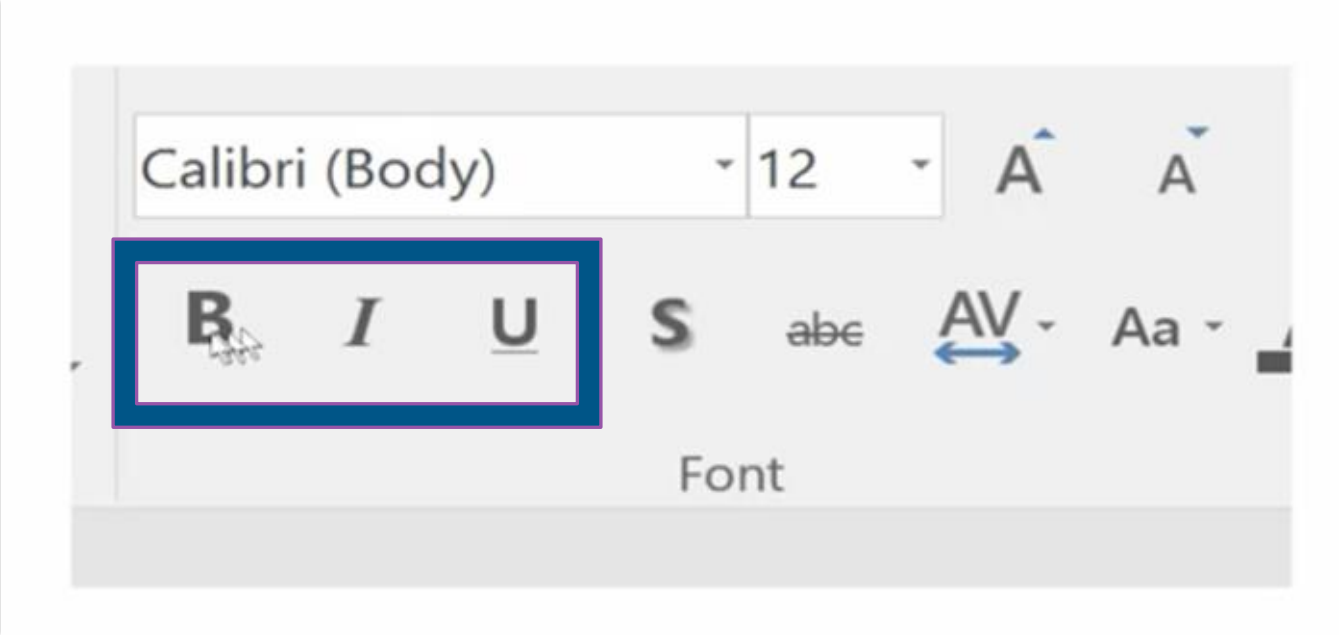
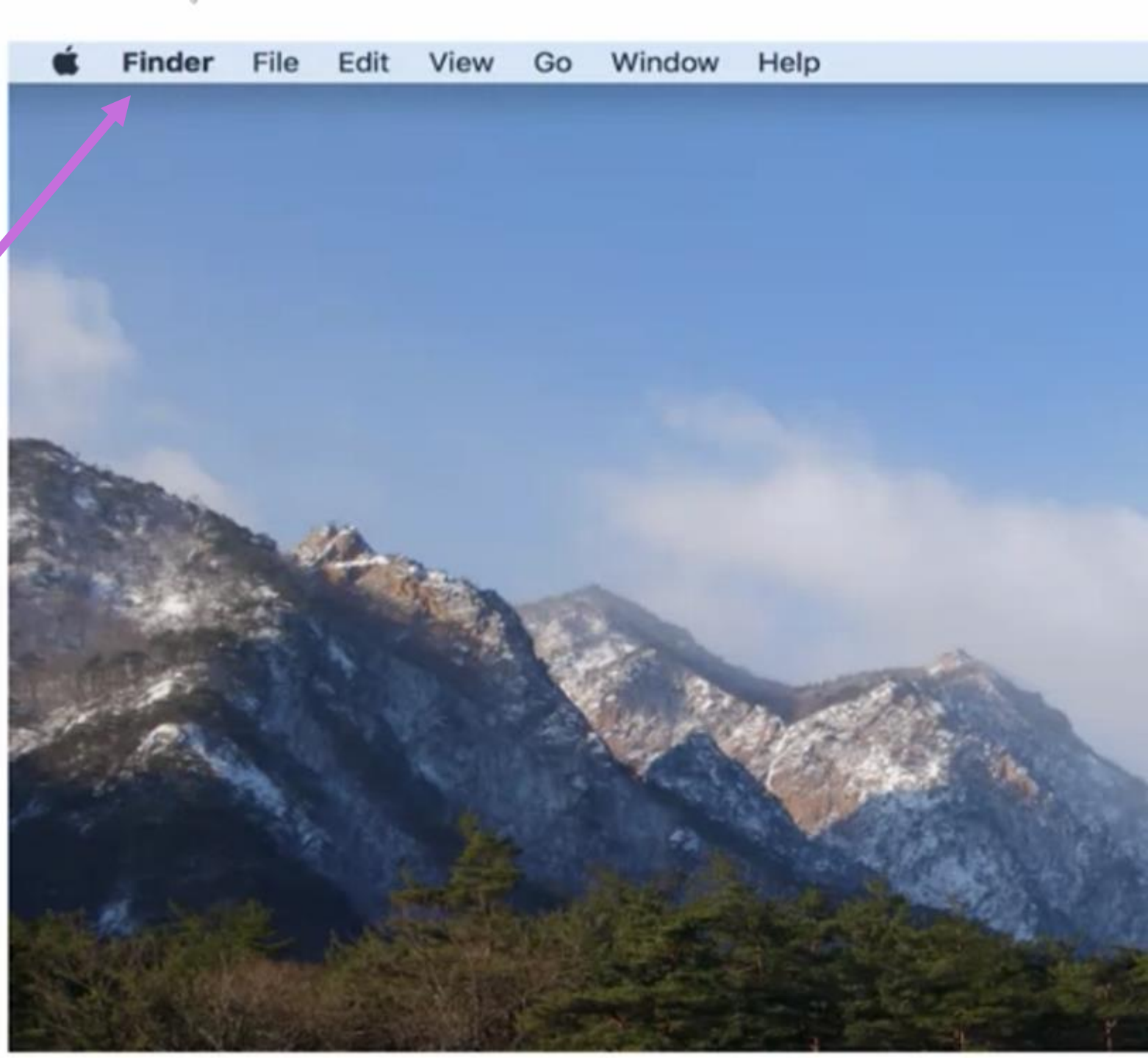
Putting Them Together

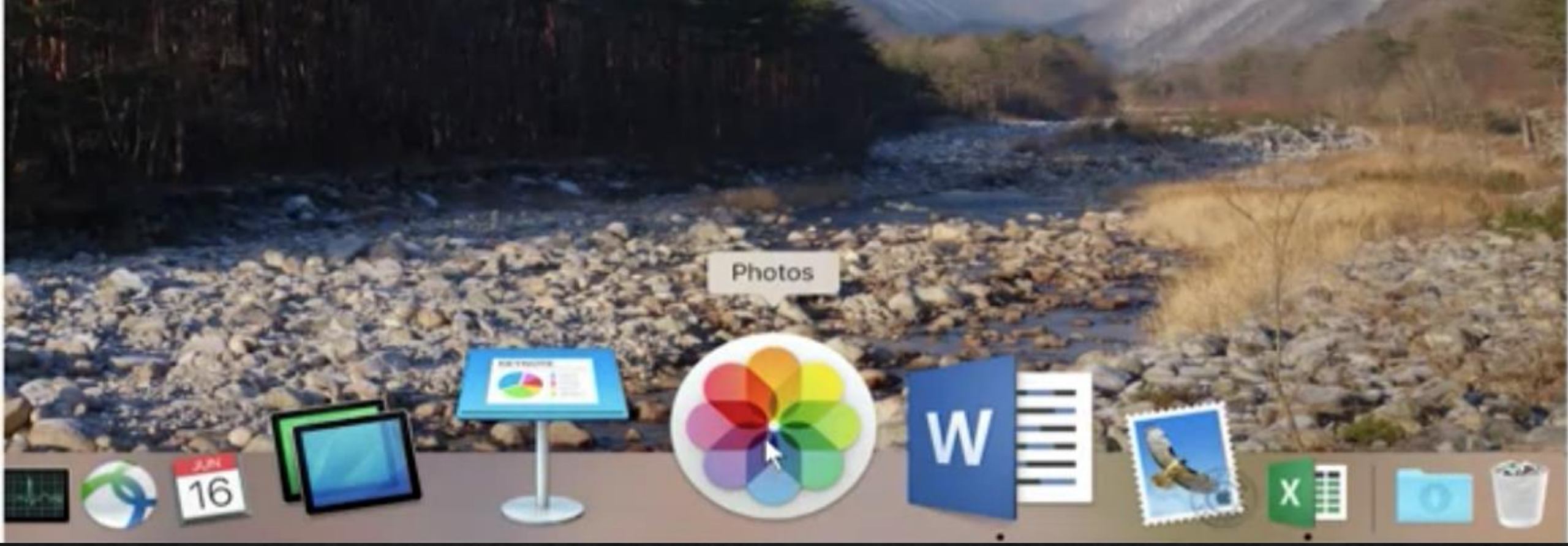


Duke

All at the Top

By placing this button at the top, it effective becomes a larger target since anywhere above it is still it





Magnifying the Target Size

Accommodating Human Characteristics

Duke

Mapping is about having a clear relationship between controls and the effect they have on the world.

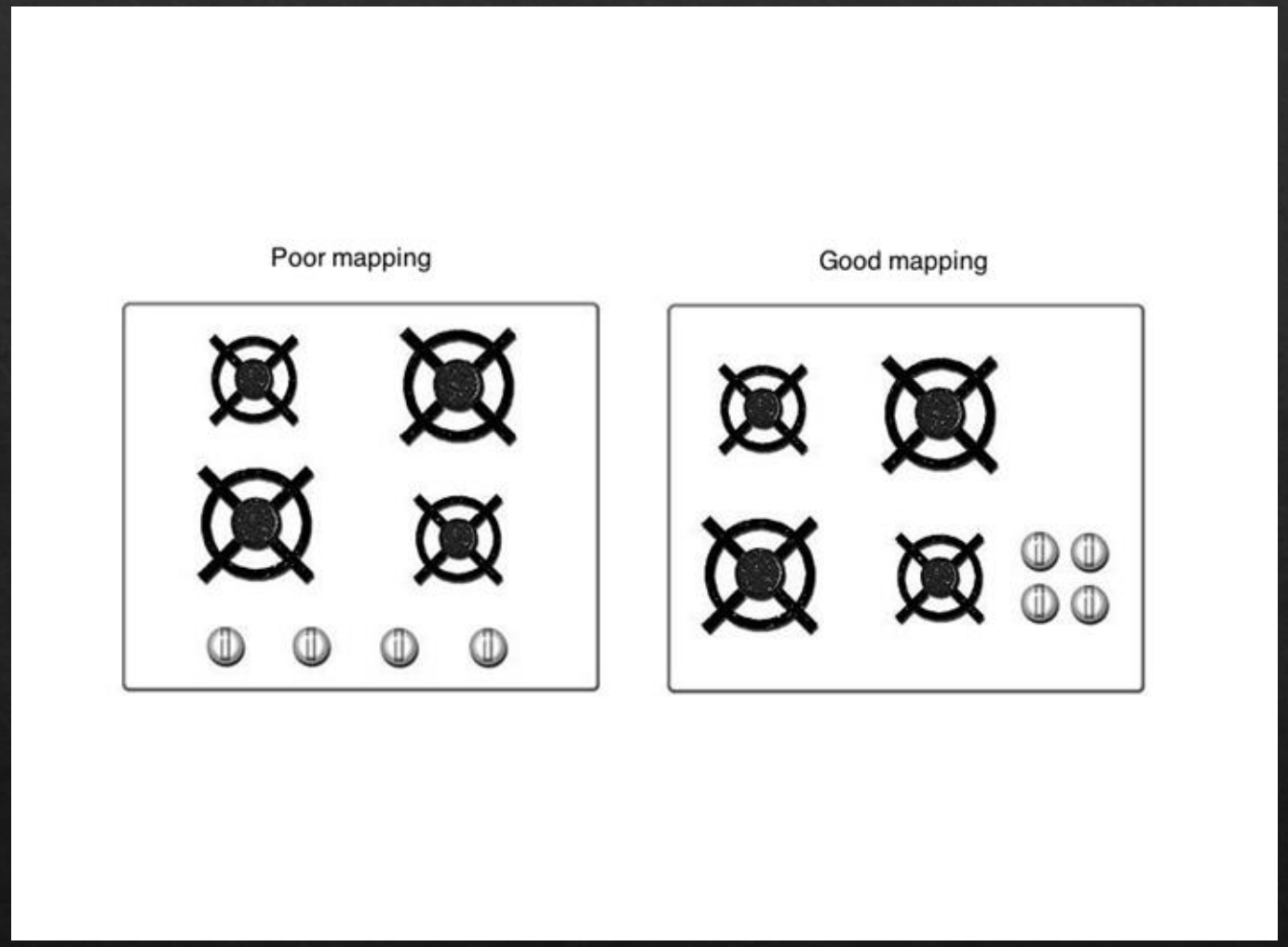
Mapping

Nestea

Refills
69¢
79¢
99¢
$1.09
$1.99

Stovetop Burners

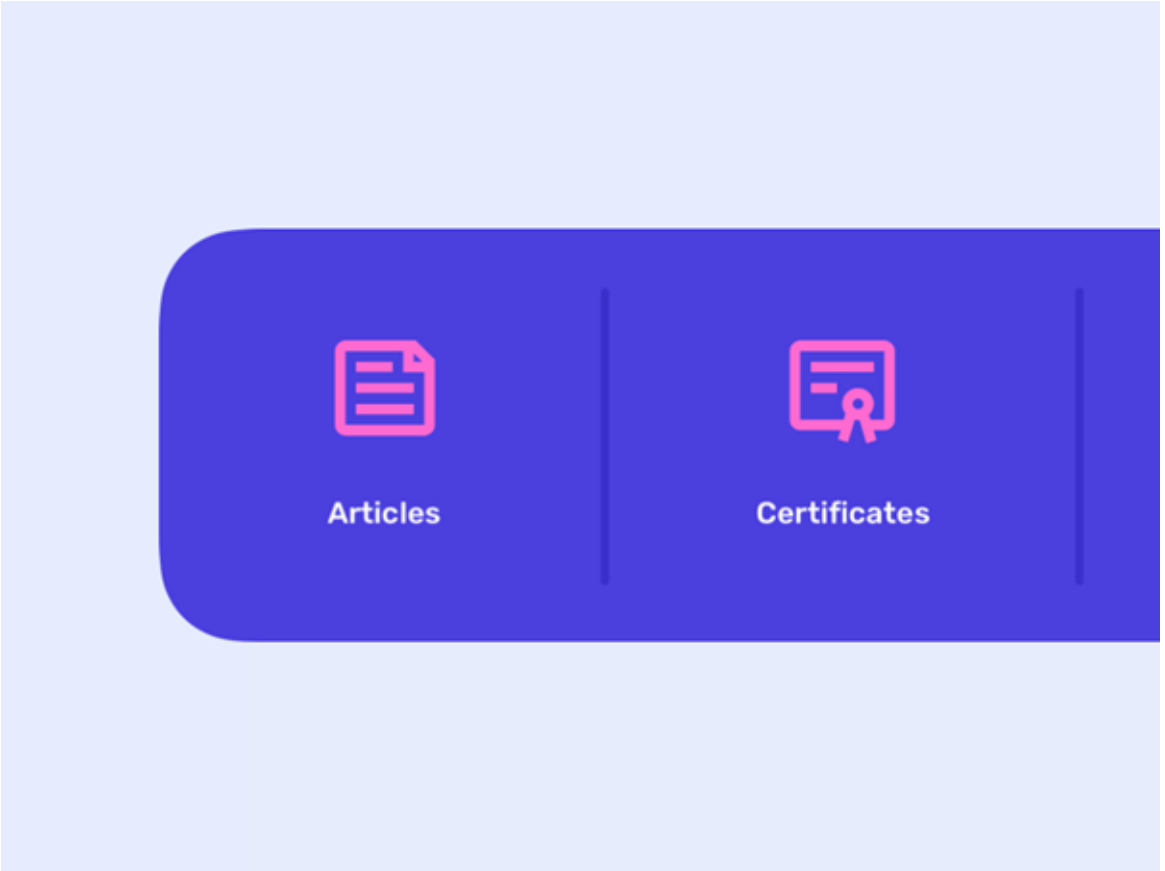
Duke



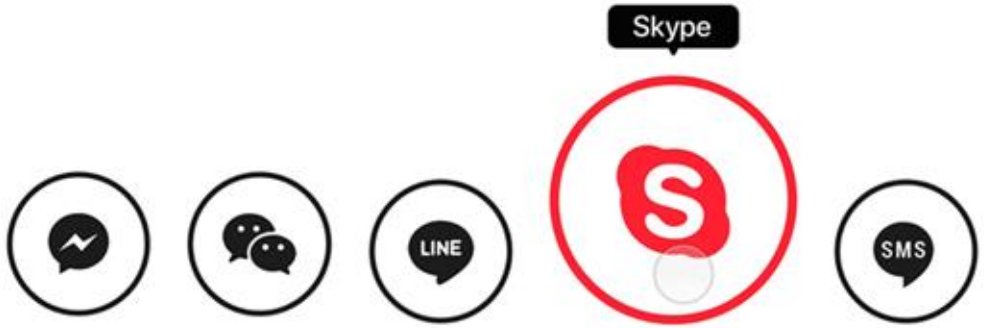
Stovetops

Feedback

Feedback is the principle of making it clear to the user what action has been taken and what has been accomplished.

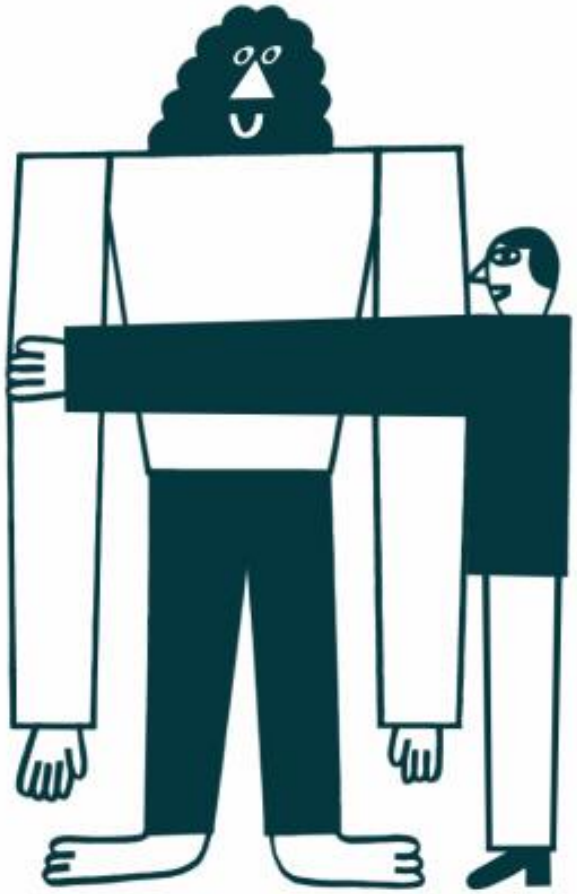


Providing relevant information specific elements



Submit

Providing relevant information about a user's action



Yikes! We're sorry for the slip up.

Please return to the previous page.

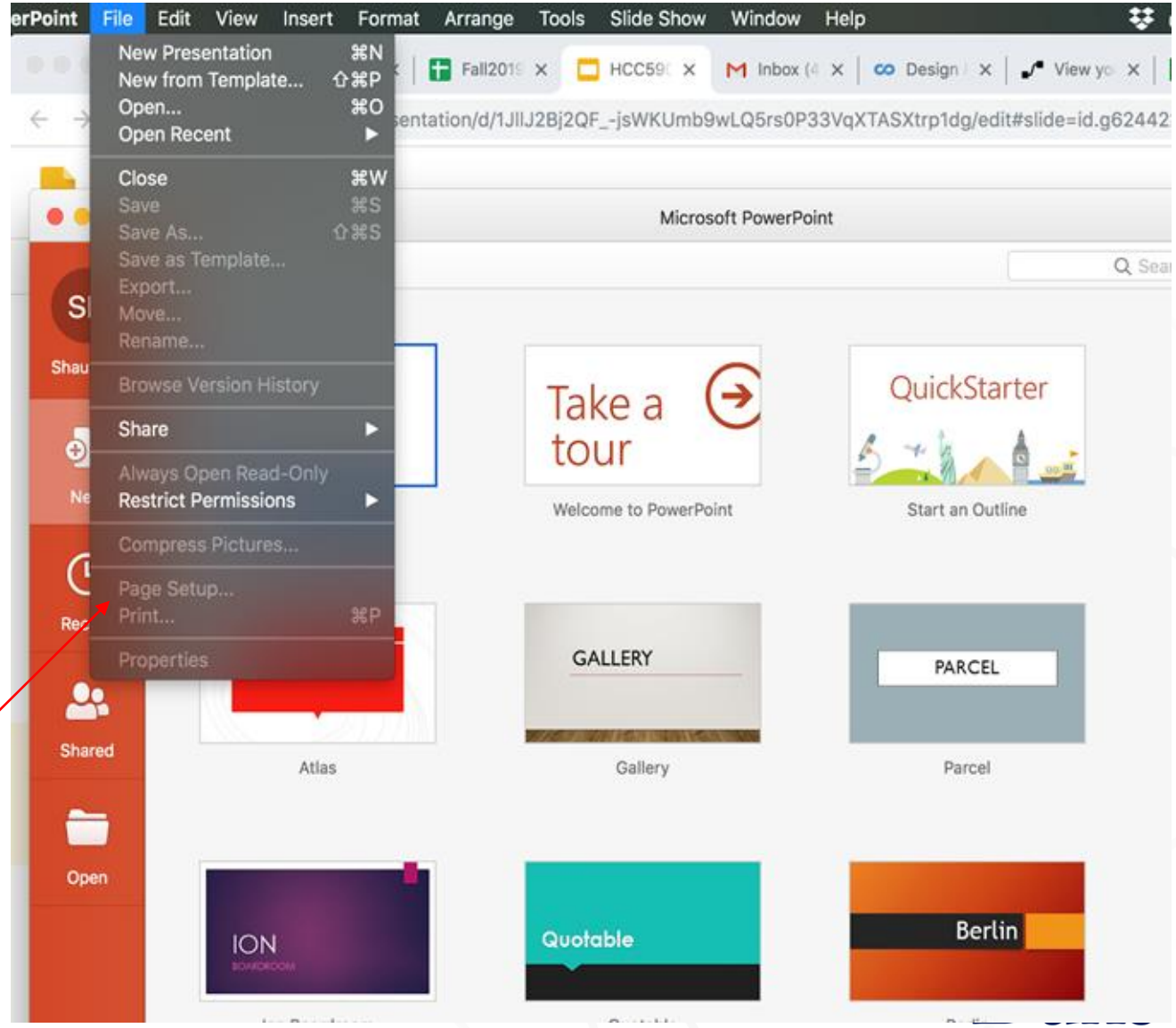
Can just be empathetic in response to user

Duke

Constraints

Duke

Grayed out for options not currently possible



Constraining user options

amazon.com/dp/0692693327?aaxitk=gvOVEVfAjrIlGi8yrV3Gog&pd_rd_i=0692693327&pf_rd_p=44fc3e0f-4b9e-4ed8-b33b-363a7257163d&h...

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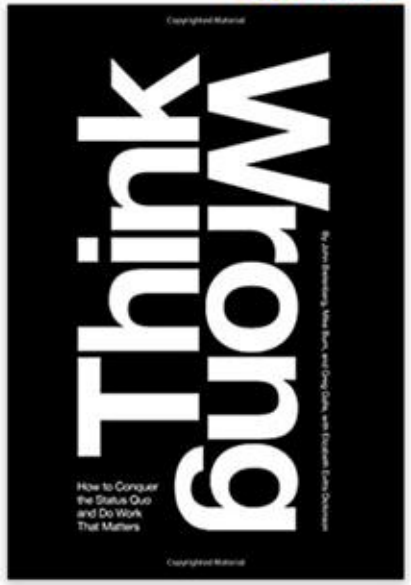

Think Wrong: How to Conquer the Status Quo and Do Work That Matters Paperback – November 1, 2016

by John Bielenberg (Author), Mike Burn (Author), Greg Galle (Author), & 1 more

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The way we solve problems is broken. We are trapped by techniques and assumptions of a prior era.

Today challenges emerge at an ever-accelerating rate, and we struggle to find the imaginative answers we crave. When we do, biology and culture conspire to obstruct our progress.

Think Wrong: How to Conquer the Status Quo and Do Work That Matters teaches you how to

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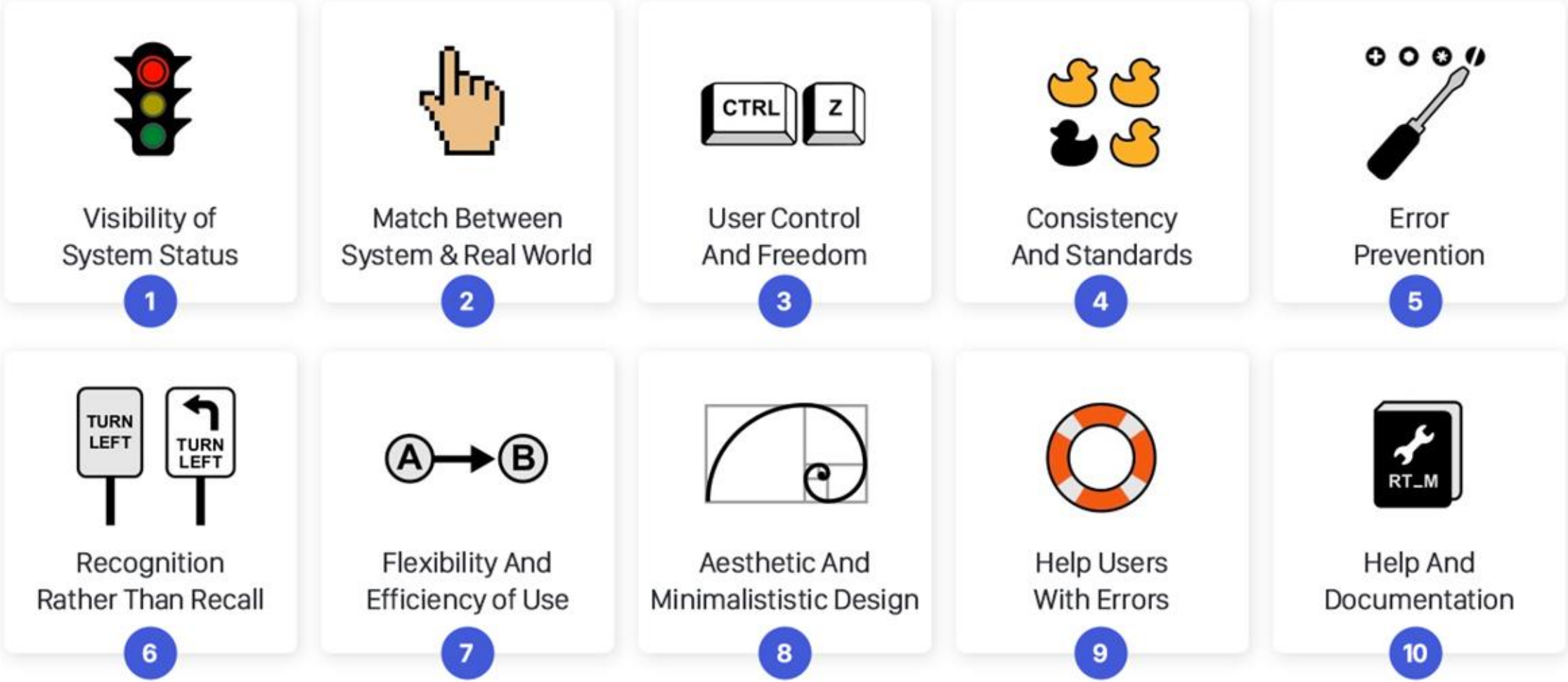
Usability Heuristics

For User Interface Design

Duke



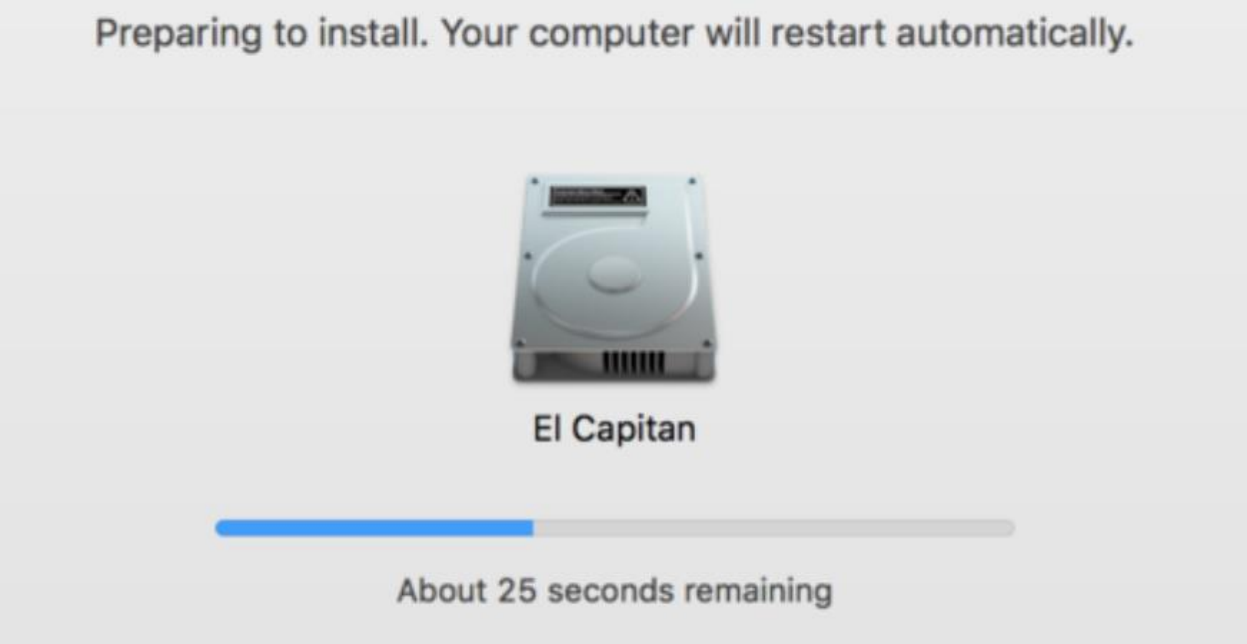
https://www.nngroup.com/articles/ten-usability-heuristics/

Visibility of System Status / Feedback

Keep users informed on what is happening

Feedback

- Relevant
- Fits importance and urgency
- Comprehensible and meaningful



Follow the user’s mental model

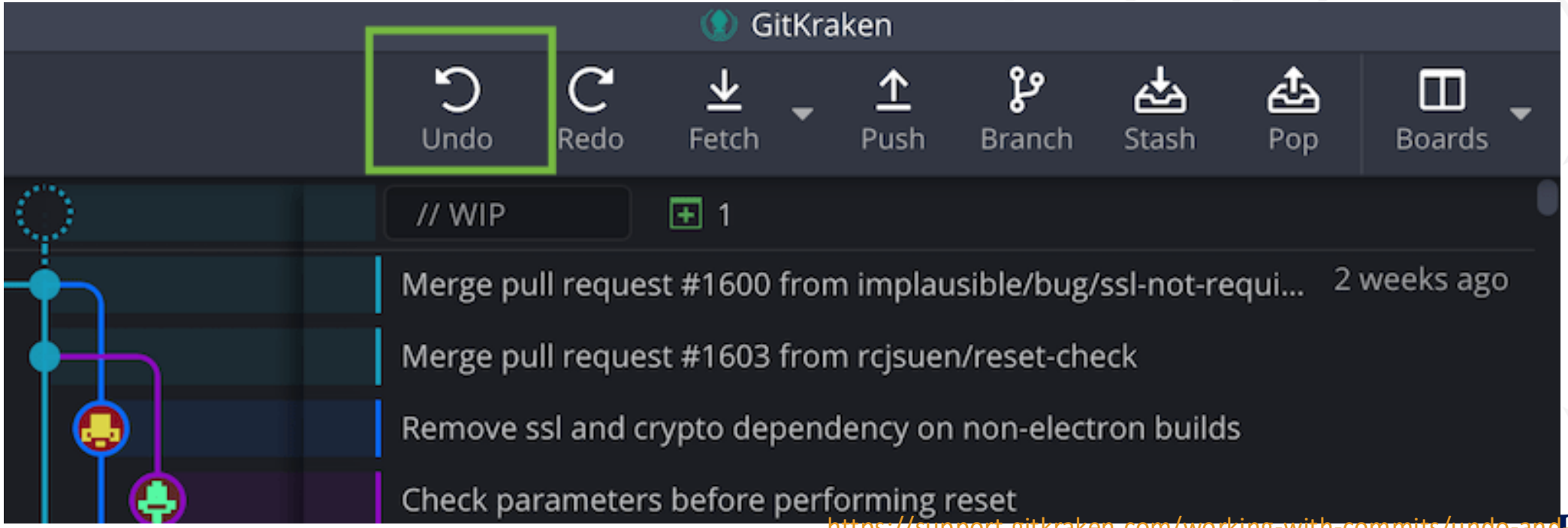
Don’t make users guess -> speak their language

Match the system and the real world

Put users in control

Let the user decide the workflow when appropriate

Permit easy reversal of actions



https://support.gitkraken.com/working-with-commits/undo-and-redo/

Consistency

Keep words and actions consistent across

- Workflow / process
- Functionality
- Appearance
- Terminology

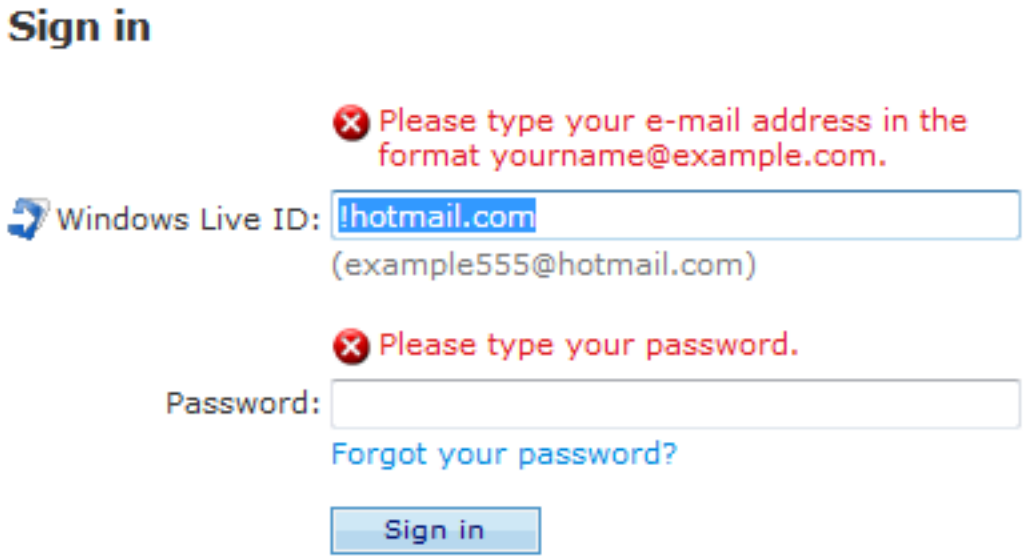
Principle of Least Surprise

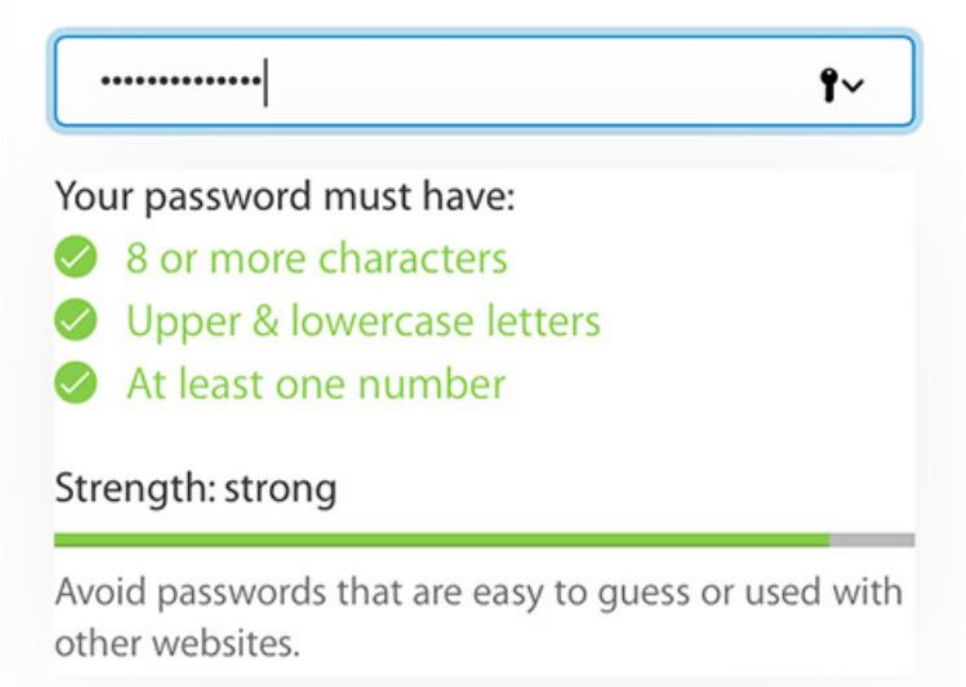
https://www.slideshare.net/littleddotnet/user-interface-design-definitions-processes-and-principles-15366100

Error Prevention and Simple Error Handling

Eliminate error-prone conditions if possible

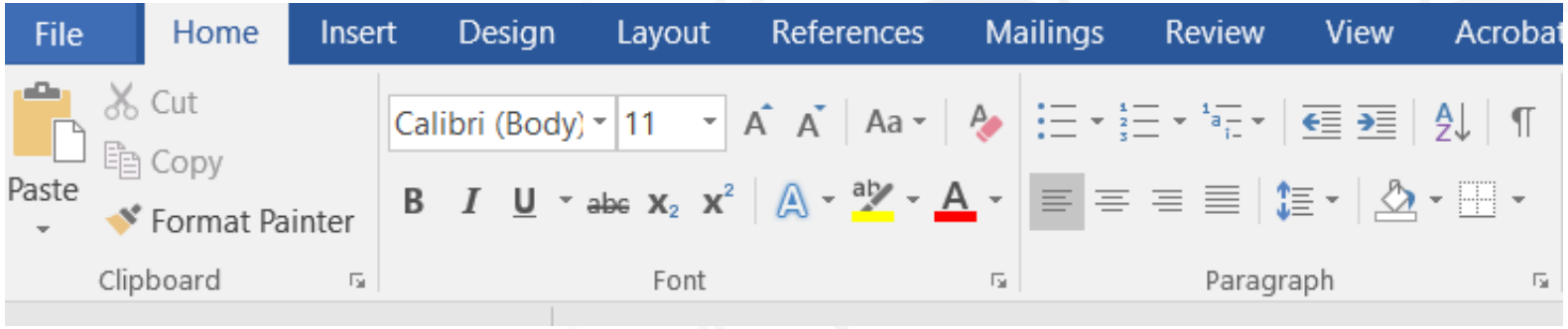
Check for errors and notify users before they commit





Reduce short term memory load

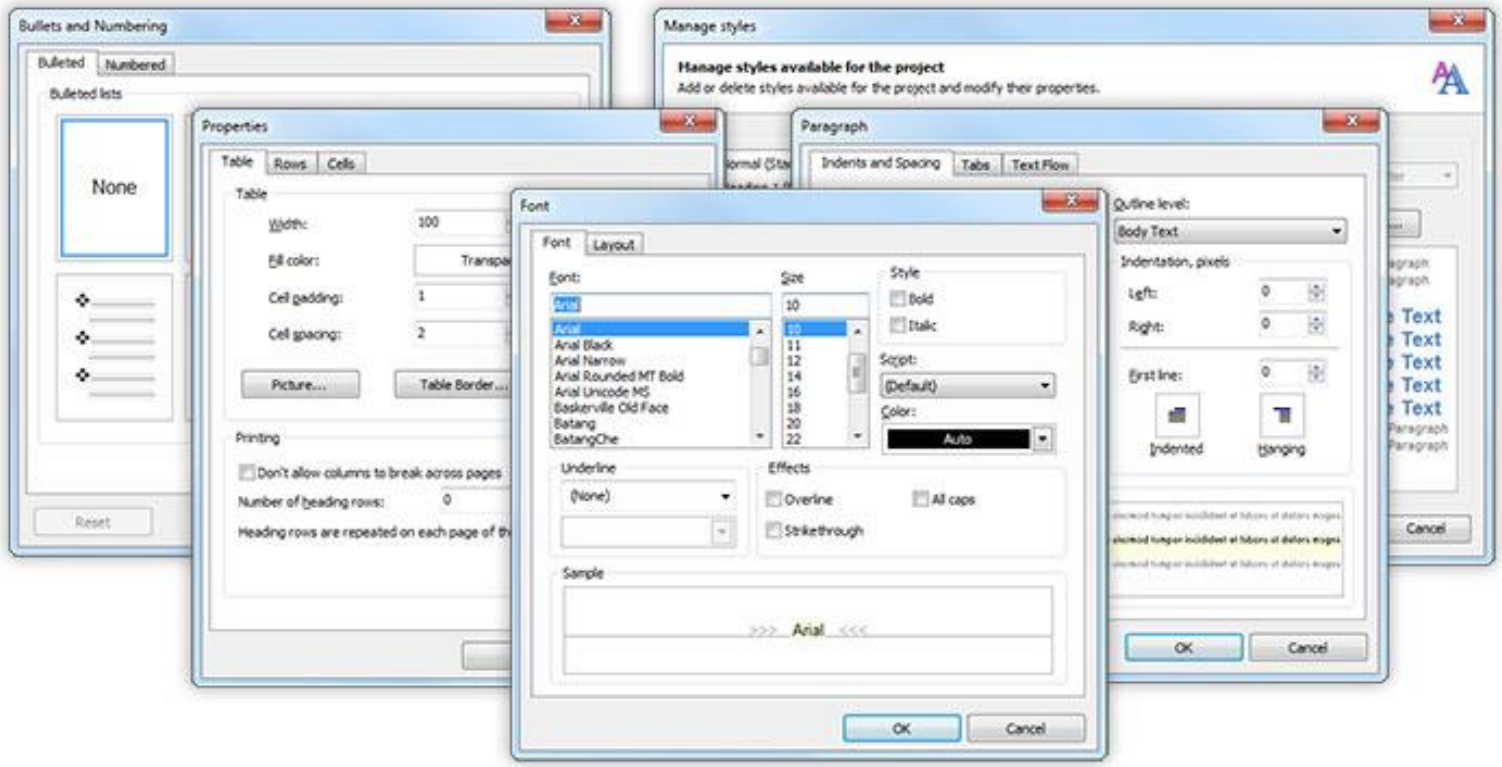
- Make objects, actions, options available
- Don’t make the user remember information
- Meaningful defaults
- Intuitive shortcuts



Recognition rather than recall

Support Power Users

- Short-cuts
- Action (macro) automatior
- Advanced features



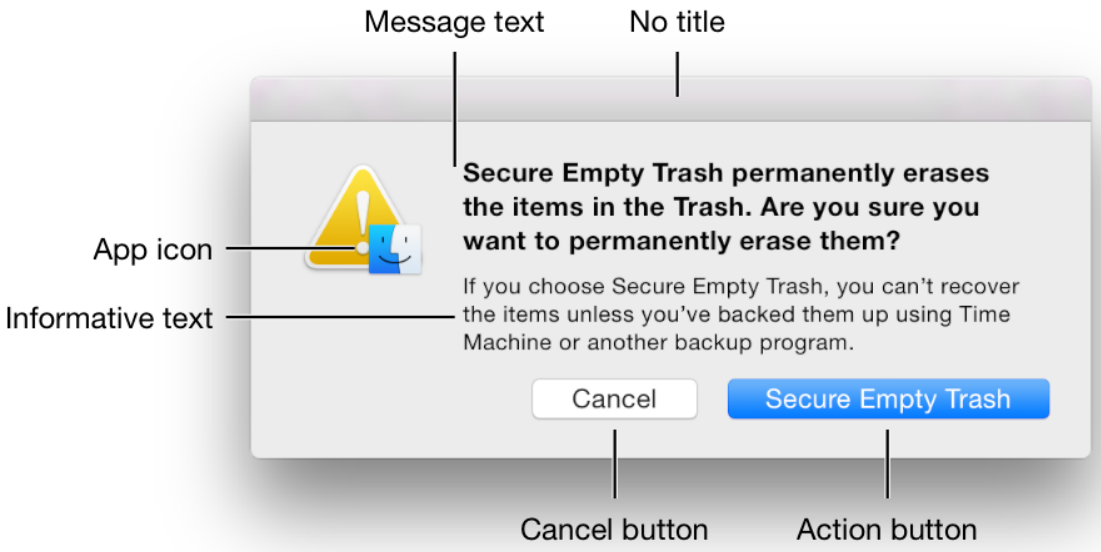
https://www.helpndoc.com/feature-tour/stunning-user-interface/

Aesthetic and Minimalist Design

All information should be valuable and relevant

Help users recognize, diagnose, and recover from errors

- Use plain language
- Precisely indicate the problem
- Suggest possible solution



Help and documentation

- Provide help and documentation
  - Focused on the user's task (context-aware)
  - List concrete steps to accomplish the task
  - Searchable

https://code.tutsplus.com/tutorials/8-design-tips-for-educating-app-users--mobile-8112

Measuring UX ('usability')

- “The main task of usability and human computer interaction testing is to evaluate how easy it is for end users to learn and to use the software.” [SWEBOK]
- “Usability inspection methods, are applied for the systematic observation of system usage under controlled conditions to determine how well people can use the system and its interfaces.
  - Examples: cognitive walkthroughs, claims analysis, field observations, thinking aloud, and even indirect approaches such as user questionnaires and interviews.” [SWEBOK]

Sit with, watch, and listen to, potential users of your product and take careful account of their feedback.

Duke

The Elements of User Experience

Jesse James Garrett
jjg@jjg.net
30 March 2000

A basic duality: The Web was originally conceived as a hypertextual information space; but the development of increasingly sophisticated front- and back-end technologies has fostered its use as a remote software interface. This dual nature has led to much confusion, as user experience practitioners have attempted to adapt their terminology to cases beyond the scope of its original application. The goal of this document is to define some of these terms within their appropriate contexts, and to clarify the underlying relationships among these various elements.

Web as software interface

Visual Design: graphic treatment of interface elements (the "look" in "look-and-feel")

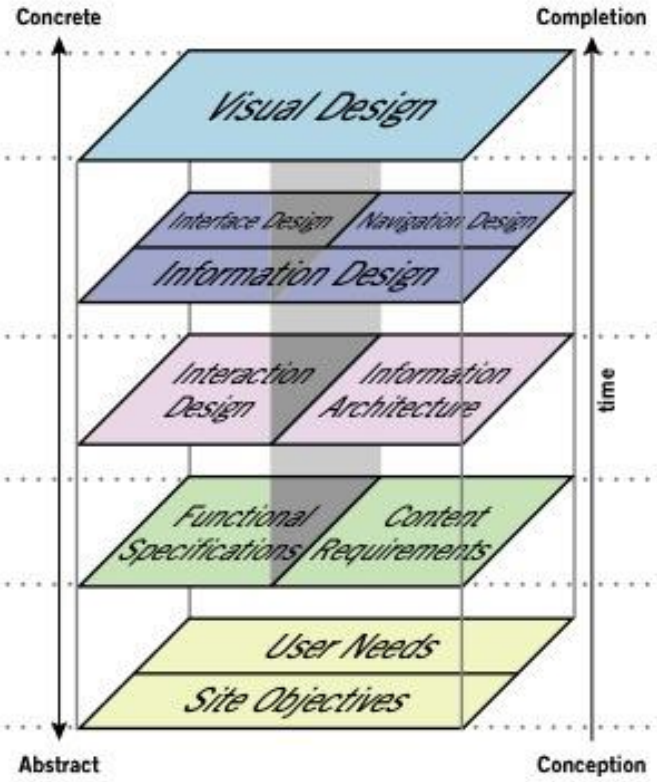
Interface Design: as in traditional HCI: design of interface elements to facilitate user interaction with functionality
Information Design: in the Tuftean sense: designing the presentation of information to facilitate understanding

Interaction Design: development of application flows to facilitate user tasks, defining how the user interacts with site functionality

Functional Specifications: "feature set": detailed descriptions of functionality the site must include in order to meet user needs

User Needs: externally derived goals for the site; identified through user research, ethno/techno/psychographics, etc.
Site Objectives: business, creative, or other internally derived goals for the site

task-oriented



Web as hypertext system

Visual Design: visual treatment of text, graphic page elements and navigational components

Navigation Design: design of interface elements to facilitate the user's movement through the information architecture
Information Design: in the Tuftean sense: designing the presentation of information to facilitate understanding

Information Architecture: structural design of the information space to facilitate intuitive access to content

Content Requirements: definition of content elements required in the site in order to meet user needs

User Needs: externally derived goals for the site; identified through user research, ethno/techno/psychographics, etc.
Site Objectives: business, creative, or other internally derived goals for the site

information-oriented

This picture is incomplete: The model outlined here does not account for secondary considerations (such as those arising during technical or content development) that may influence decisions during user experience development. Also, this model does not describe a development process, nor does it define roles within a user experience development team. Rather, it seeks to define the key considerations that go into the development of user experience on the Web today.

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http://www.jjg.net/ia/

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